# DRIVELINE/AXLE

**03** SECTION



## 03-00 OUTLINE



### DRIVELINE/AXLE ABBREVIATIONS

dcf030000000t01

| | |
|---|---|
| ABS | Antilock Brake System |
| API | American Petroleum Institute |
| DLC-2 | Data Link Connector-2 |
| DTC | Diagnostic Trouble Code |
| LSD | Limited Slip Differential |
| OFF | Switch Off |
| ON | Switch On |
| PCM | Powertrain Control Module |
| RFW | Remote Freewheel |
| SAE | Society of Automotive Engineers |
| 4x2 | 4-Wheel 2-Drive |
| 4x4 | 4-Wheel 4-Drive |

### DRIVELINE/AXLE FEATURES

dcf030000000t02

| | |
|---|---|
| Improved rigidity, reduced noise and vibration | • Constant velocity joint type rear drive shaft adopted<br>• 2-part, 3-joint type propeller shaft with a center bearing support has been adopted (4×2, 4×4 rear)<br>• 1-part type propeller shaft has been adapted (4×4 front)<br>• Double offset-shaped constant velocity joint adopted for differential-side joint of front drive shaft<br>• Bell-shaped constant velocity joint adopted for wheel-side joint of front drive shaft |
| Improved durability | • Unit-design taper roller bearing has been adopted |
| Improved reliability | • A spring-tensioned clip-type method of fixing the universal joint has been adopted (4×2, 4×4 rear) |
| Improved operability | • A direct control type transfer shift lever adopted<br>• Remote freewheel (RFW) has been adopted<br>• Electronic control type transfer case adopted |
| Improved serviceability | • Unit-design taper roller bearing has been adopted |
| Improved driveability | • LSD has been adopted |

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## DRIVELINE/AXLE SPECIFICATIONS

dcf030000000t03

<table>
<tr><th>Item</th><th colspan="5">Specification</th></tr>
<tr><td>Engine</td><td>WL-3</td><td colspan="2">WL-C</td><td colspan="2">WE-C</td></tr>
<tr><td>Transmission type</td><td>R15M-D</td><td colspan="3">S15M-D</td><td>S15MX-D</td></tr>
<tr><td>Vehicle type</td><td>4×2</td><td>4×2</td><td colspan="3">4×2 Hi-Rider</td><td>4×4</td></tr>
<tr><td colspan="6">Front axle</td></tr>
<tr><td>Bearing type</td><td colspan="5">Taper roller bearing</td></tr>
<tr><td colspan="6">Rear axle</td></tr>
<tr><td>Bearing type</td><td colspan="5">Taper roller bearing</td></tr>
<tr><td>Support type</td><td colspan="5">Semi-floating</td></tr>
<tr><td>Casing</td><td colspan="5">Banjo type</td></tr>
<tr><td>Length (mm {in})</td><td colspan="5">739 {29.1}</td></tr>
<tr><td>Diameter (mm {in})</td><td colspan="5">35.0 {1.4}</td></tr>
<tr><td colspan="6">Rear differential</td></tr>
<tr><td>Reduction gear</td><td colspan="5">Hypoid gear</td></tr>
<tr><td>Differential gear</td><td colspan="5">Straight bevel gear</td></tr>
<tr><td>Ring gear size (Inches)</td><td>8.0</td><td colspan="4">8.9</td></tr>
<tr><td>Final gear ratio</td><td>3.909</td><td>3.416</td><td colspan="3">3.727</td></tr>
<tr><td>Differential oil — Type — Grade</td><td colspan="5">API service GL-5</td></tr>
<tr><td>Differential oil — Type — Viscosity</td><td colspan="5">SAE 90</td></tr>
<tr><td>Differential oil — Type — Capacity (L {US qt, Imp qt})</td><td>1.40 {1.48, 1.23}</td><td>2.45 {2.32, 2.04}</td><td colspan="3">2.35 {2.22, 1.96}</td></tr>
<tr><td colspan="6">Front differential</td></tr>
<tr><td>Reduction gear</td><td colspan="4">–</td><td>Hypoid gear</td></tr>
<tr><td>Differential gear</td><td colspan="4">–</td><td>Straight bevel gear</td></tr>
<tr><td>Ring gear size (Inches)</td><td colspan="4">–</td><td>8.00</td></tr>
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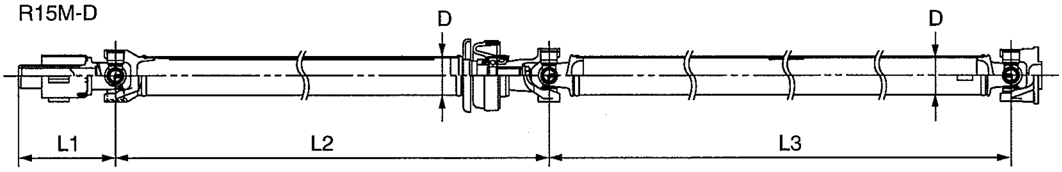
<tr><td>Differential oil — Type — Viscosity</td><td colspan="4">–</td><td>Above -18 °C {0 °F}: SAE90<br>Below -18 °C {0 °F}: SAE80</td></tr>
<tr><td>Differential oil — Type — Capacity (L {US qt, Imp qt})</td><td colspan="4">–<br>–</td><td>1.9 {1.8, 1.6}</td></tr>
<tr><td colspan="6">Front drive shaft</td></tr>
<tr><td>Joint type — Wheel side</td><td colspan="4">–</td><td>Bell joint</td></tr>
<tr><td>Joint type — Differential side</td><td colspan="4">–</td><td>Double offset joint</td></tr>
<tr><td>Shaft diameter (mm {in})</td><td colspan="4">–</td><td>30.0 {1.18}</td></tr>
<tr><td colspan="6">Front propeller shaft</td></tr>
<tr><td>Length (front) (mm {in}) L1</td><td colspan="4">–</td><td>45.3 {1.78}</td></tr>
<tr><td>Length (front) (mm {in}) L2</td><td colspan="4">–</td><td>552.1 {21.74}</td></tr>
<tr><td>Length (front) (mm {in}) L3</td><td colspan="4">–</td><td>45.3 {1.78}</td></tr>
<tr><td>Outer diameter (front) (mm {in}) D</td><td colspan="4">–</td><td>63.5 {2.50}</td></tr>
<tr><td colspan="6">Rear propeller shaft</td></tr>
<tr><td>Length (rear) (mm {in}) L1</td><td>159 {6.26}</td><td colspan="3">163.4 {6.433}</td><td>41.1 {1.62}</td></tr>
<tr><td>Length (rear) (mm {in}) L2</td><td>666.1 {26.22}</td><td colspan="3">670.6 {26.40}</td><td>443.6 {17.46}</td></tr>
<tr><td>Length (rear) (mm {in}) L3</td><td>956.1 {37.64}</td><td>909.8 {35.82}</td><td colspan="2">890.3 {35.05}</td><td>896.3 {35.29}</td></tr>
<tr><td>Outer diameter (rear) (mm {in}) D</td><td colspan="5">63.5 {2.50}</td></tr>
<tr><td>Joint type</td><td colspan="5">Cross-shaped joint</td></tr>
</table>

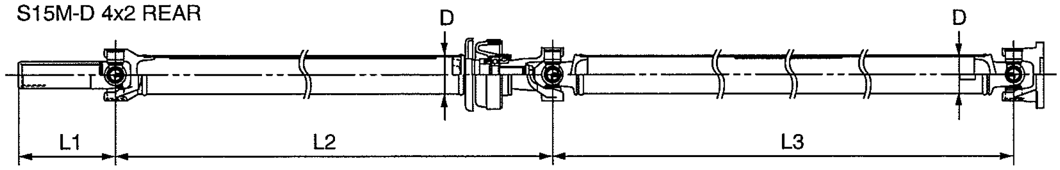
# OUTLINE

| Item | Specification | | | | |
|---|---|---|---|---|---|
| Engine | WL-3 | WL-C | | WE-C | |
| Transmission type | R15M-D | S15M-D | | | S15MX-D |
| Vehicle type | 4×2 | 4×2 | 4×2 Hi-Rider | | 4×4 |

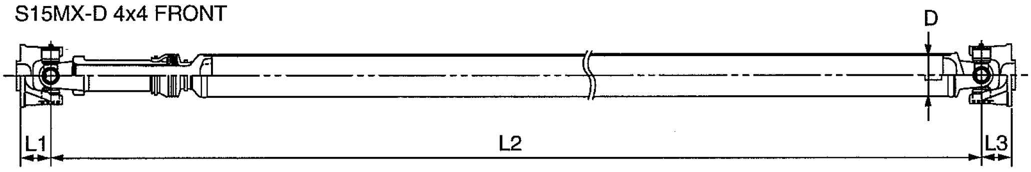
R15M-D



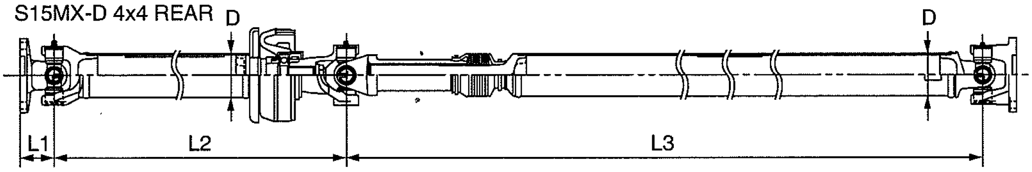

S15M-D 4x2 REAR



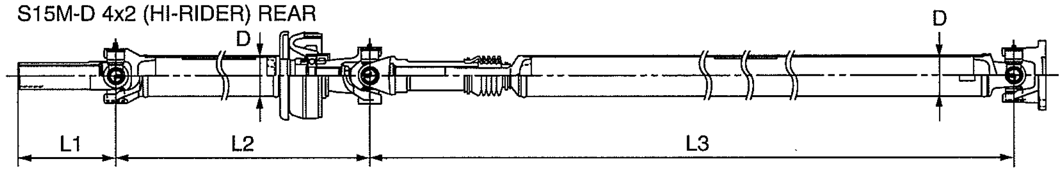

S15MX-D 4x4 FRONT




S15MX-D 4x4 REAR




S15M-D 4x2 (HI-RIDER) REAR

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<table>
<tr><th colspan="3">Item</th><th colspan="5">Specification</th></tr>
<tr><td colspan="3">Engine</td><td>WL-3</td><td colspan="2">WL-C</td><td colspan="2">WE-C</td></tr>
<tr><td colspan="3">Transmission type</td><td>R15M-D</td><td colspan="3">S15M-D</td><td>S15MX-D</td></tr>
<tr><td colspan="3">Vehicle type</td><td>4×2</td><td>4×2</td><td colspan="2">4×2 Hi-Rider</td><td>4×4</td></tr>
<tr><td colspan="8">Transfer</td></tr>
<tr><td rowspan="2">Transfer oil</td><td colspan="2">Type</td><td colspan="4">–</td><td>Mercon® Multi-purpose AFT XT-2-QDX</td></tr>
<tr><td>Oil capacity (approx. quantity)</td><td>(L {US qt, Imp qt})</td><td colspan="4">–</td><td>1.85 {1.95, 1.63}</td></tr>
</table>

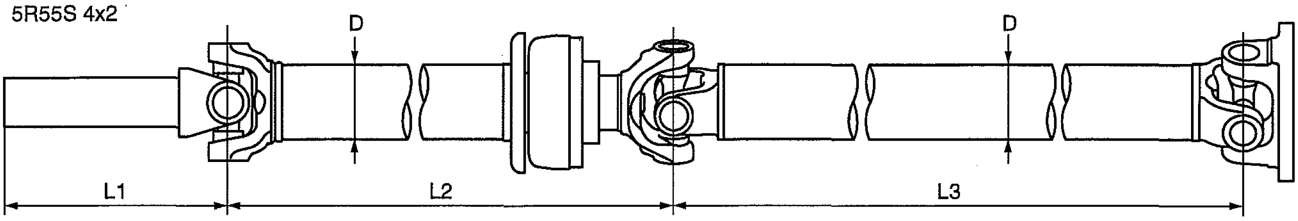
## DRIVELINE/AXLE SPECIFICATIONS [RANGER (5R55S)]

id030000100211

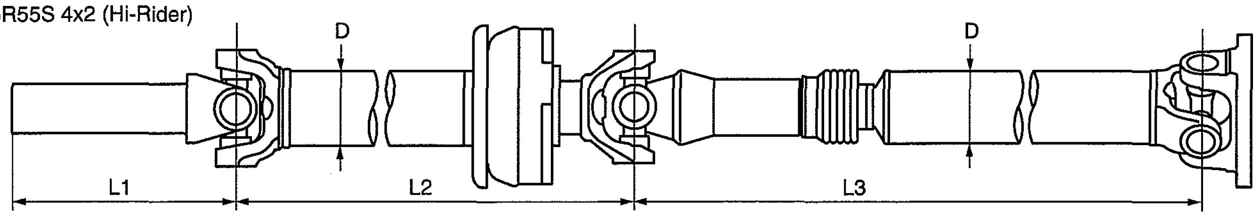
| Item | | | Specification | | |
|---|---|---|---|---|---|
| **Engine** | | | **WL-C** | **WE-C** | |
| **Transmission type** | | | **5R55S** | | |
| **Vehicle type** | | | **4×2** | **4×2 Hi-Rider** | **4×4** |
| **Front axle** | | | | | |
| Bearing type | | | Taper roller bearing | | |
| **Rear axle** | | | | | |
| Bearing type | | | Taper roller bearing | | |
| Support type | | | Semi-floating | | |
| Casing | | | Banjo type | | |
| Length (mm {in}) | | | 739 {29.1} | | |
| Diameter (mm {in}) | | | 35.0 {1.4} | | |
| **Rear differential** | | | | | |
| Reduction gear | | | Hypoid gear | | |
| Differential gear | | | Straight bevel gear | | |
| Ring gear size (Inches) | | | 8.9 | | |
| Final gear ratio | | | 3.416 | 3.727 | |
| Differential oil | Type | Grade | API service GL-5 | | |
| | | Viscosity | SAE 90 | | |
| | | Capacity (approx. quantity) (L {US qt, Imp qt}) | 2.45 {2.32, 2.04} | 2.35 {2.22, 1.96} | |
| **Front differential** | | | | | |
| Reduction gear | | | – | | Hypoid gear |
| Differential gear | | | – | | Straight bevel gear |
| Ring gear size (Inches) | | | – | | 8.00 |
| Final gear ratio | | | – | | 3.727 |
| Differential oil | Type | Grade | – | | API service GL-5 |
| | | Viscosity | – | | Above -18 °C {0 °F}: SAE90<br>Below -18 °C {0 °F}: SAE80 |
| | | Capacity (approx. quantity) (L {US qt, Imp qt}) | – | | 1.9 {1.8, 1.6} |
| **Front drive shaft** | | | | | |
| Joint type | Wheel side | | – | | Bell joint |
| | Differential side | | – | | Double offset joint |
| Shaft diameter | (mm {in}) | | – | | 30.0 {1.18} |
| **Front propeller shaft** | | | | | |
| Length (front) | (mm {in}) | L1 | – | | 13.2 {0.52} |
| | | L2 | – | | 588.4 {23.17} |
| | | L3 | – | | 34.3 {1.35} |
| Outer diameter (front) | (mm {in}) | D | – | | 26.5 {1.04} |
| **Rear propeller shaft** | | | | | |
| Length (rear) | (mm {in}) | L1 | 192.5 {7.579} | | 41.1 {1.62} |
| | | L2 | 677.6 {26.68} | 691.6 {27.23} | 443.6 {17.46} |
| | | L3 | 909.8 {35.82} | 890.3 {35.05} | 896.3 {35.29} |
| Outer diameter (rear) | (mm {in}) | D | 63.5 {2.50} | | |
| Joint type | | | Cross-shaped joint | | |

| Item | Specification | | |
|---|---|---|---|
| **Engine** | **WL-C** | **WE-C** | |
| **Transmission type** | **5R55S** | | |
| **Vehicle type** | **4×2** | **4×2 Hi-Rider** | **4×4** |

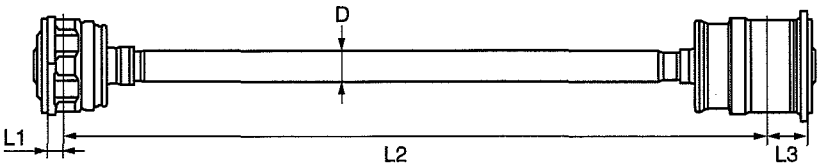
5R55S 4x2



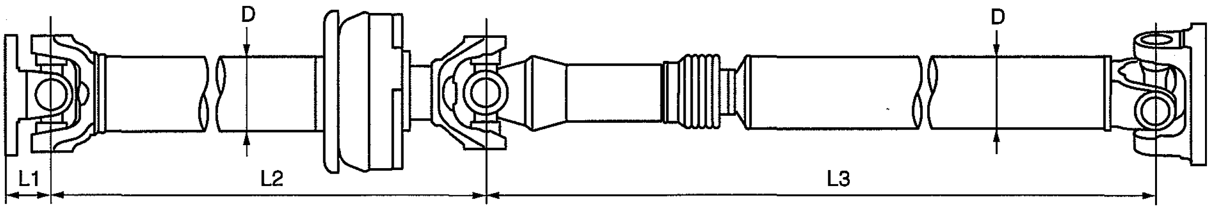

5R55S 4x2 (Hi-Rider)




5R55S 4x4 FRONT




5R55S 4x4 REAR




| **Transfer case** | | | |
|---|---|---|---|
| Gear ratio | High | – | 1.000 |
| | Low | – | 2.480 |
| Transfer case oil | Type | – | Mercon® V |
| | Oil capacity (approx. quantity) (L {US qt, Imp qt}) | – | 1.2 {1.3, 1.1} |

# 03-02 ON-BOARD DIAGNOSTIC

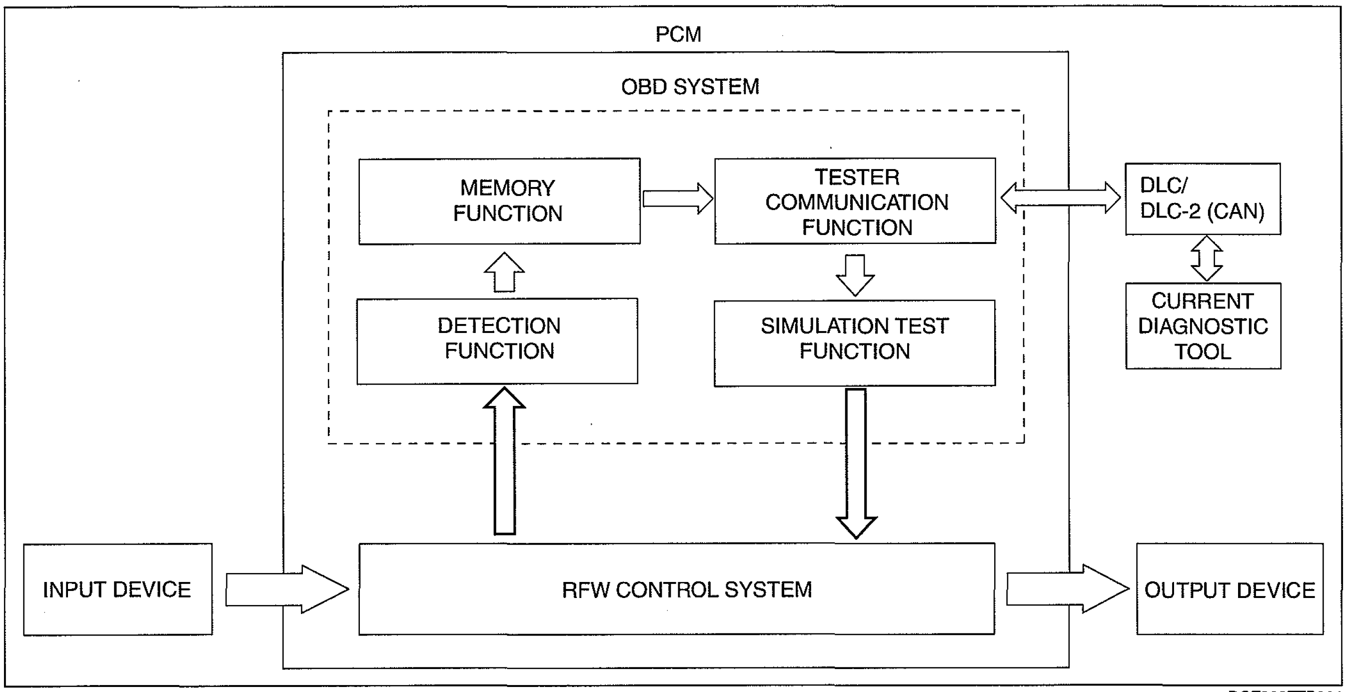


## ON-BOARD DIAGNOSTIC OUTLINE [REMOTE FREEWHEEL (RFW)]

dcf030227100t01

- The on-board diagnostic function allows for detecting malfunctions in the input/output signals when the engine switch is at the ON position.
- The DLC/DLC-2, which combines the failure detection and detection maintenance connectors, has been adopted to improve serviceability. By connecting a current diagnostic tool to the DLC/DLC-2, malfunction diagnosis can be carried out.
- Using a current diagnostic tool, DTCs can be retrieved or erased depending on the screen display, thus improving serviceability.
- The on-board diagnostic function of remote freewheel (RFW) is controlled by PCM.

### Block Diagram




DCF302ZTB001

### DTC Table (Related to the RFW)

| DTC | Malfunction location | DTC stored in memory |
|---|---|---|
| P1812 | RFW Lock solenoid valve circuit failure | X |
| P1813 | RFW Lock solenoid valve open circuit | X |
| P1814 | RFW Lock solenoid valve circuit/short to battery | X |
| P1815 | RFW Lock solenoid valve circuit/short to ground | X |
| P1878 | RFW Free solenoid valve circuit failure | X |
| P1879 | RFW Free solenoid valve open circuit | X |
| P1880 | RFW Free solenoid valve circuit/short to battery | X |
| P1885 | RFW Free solenoid valve circuit/short to ground | X |

### Simulation Item Table (Related to the RFW)

| Item | Output part | Operation | Test condition |
|---|---|---|---|
| 4WDMODE_L | 4x4 indicator light | Off/On | Engine switch at ON |
| HUBLOCK | RFW lock solenoid valve | Off/On | |
| HUBLOCK_L | RFW free solenoid valve | Off/On | |
| NTFLAMP | RFW indicator light | Off/On | |

# 03-02 ON-BOARD DIAGNOSTIC



## ON-BOARD DIAGNOSTIC OUTLINE [4x4 control module]

id0302a1100100

### Special Features

- When a 4x4 control system malfunction is detected, the 4x4 indicator light and 4L indicator light are flashed to notify the occurrence of the malfunction.

### Memory Function

- The memory function records signal systems which have been determined to be abnormal by the malfunction detection function, and the recorded malfunction information is not cleared even when the engine switch is turned off (LOCK position) or the malfunction is repaired.
- Refer to the Workshop Manual for the DTC clearing procedure.

### Malfunction Indication Function

- The malfunction indication function flashes the 4x4 indicator light and 4L indicator light as a DTC based on the signal determined to be a malfunction by the malfunction detection function.

### DTC Table

| DTC No. | Malfunction location | DTC stored in memory |
|---|---|---|
| 1 | 4x4 control module malfunction | X |
| 2 | Motor component (shift motor) malfunction | X |
| 3 | Clutch coil malfunction | X |
| 4 | Speed sensor malfunction | X |
| 5 | Solenoid valve malfunction | X |
| 6 | 4x4 switch malfunction | X |
| 7 | Motor component (position sensor) malfunction | X |

# 03-11 FRONT AXLE



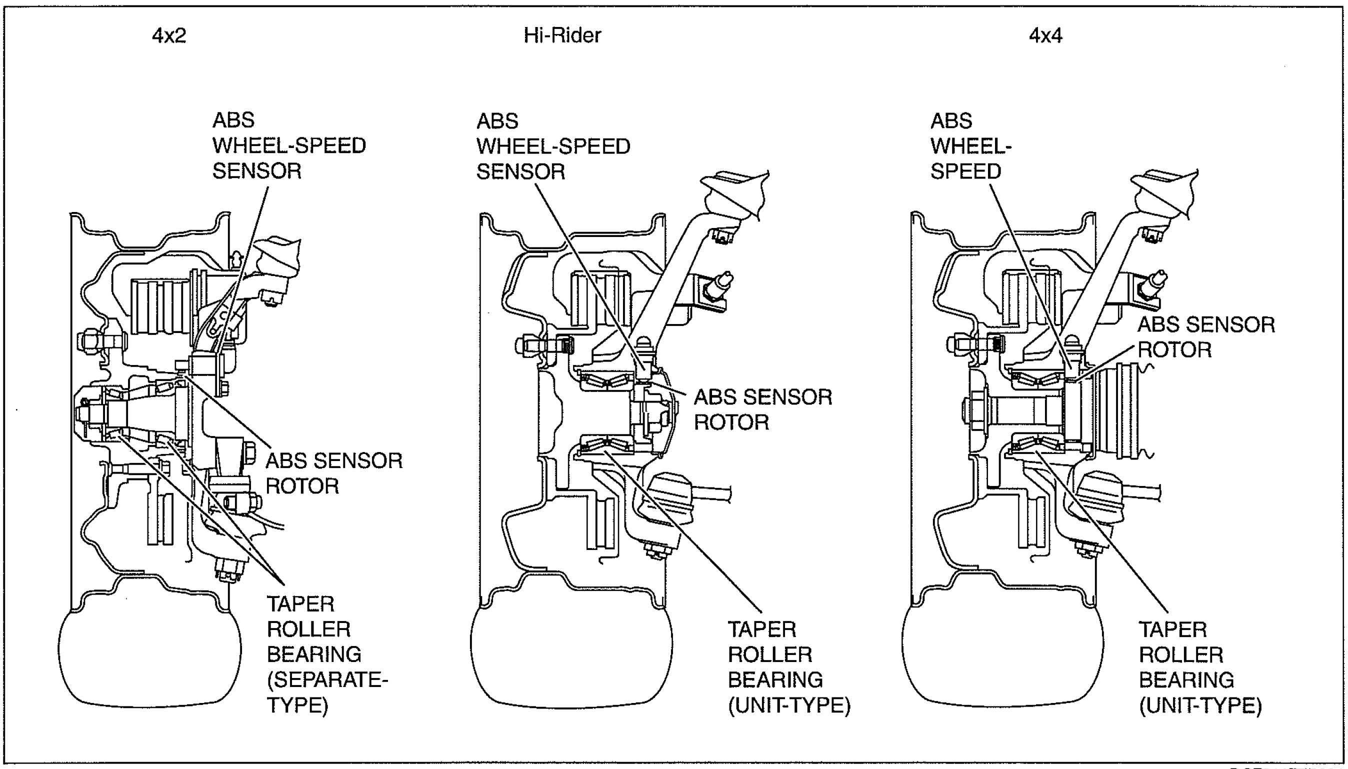
## FRONT AXLE OUTLINE

dcf031104000t01

- Taper roller bearings, which have a large load capacity, are used as wheel bearings, thus preload adjustment is necessary.

## FRONT AXLE CROSS-SECTIONAL VIEW

dcf031104000t02




DCF311ZTB001

# 03-12 REAR AXLE



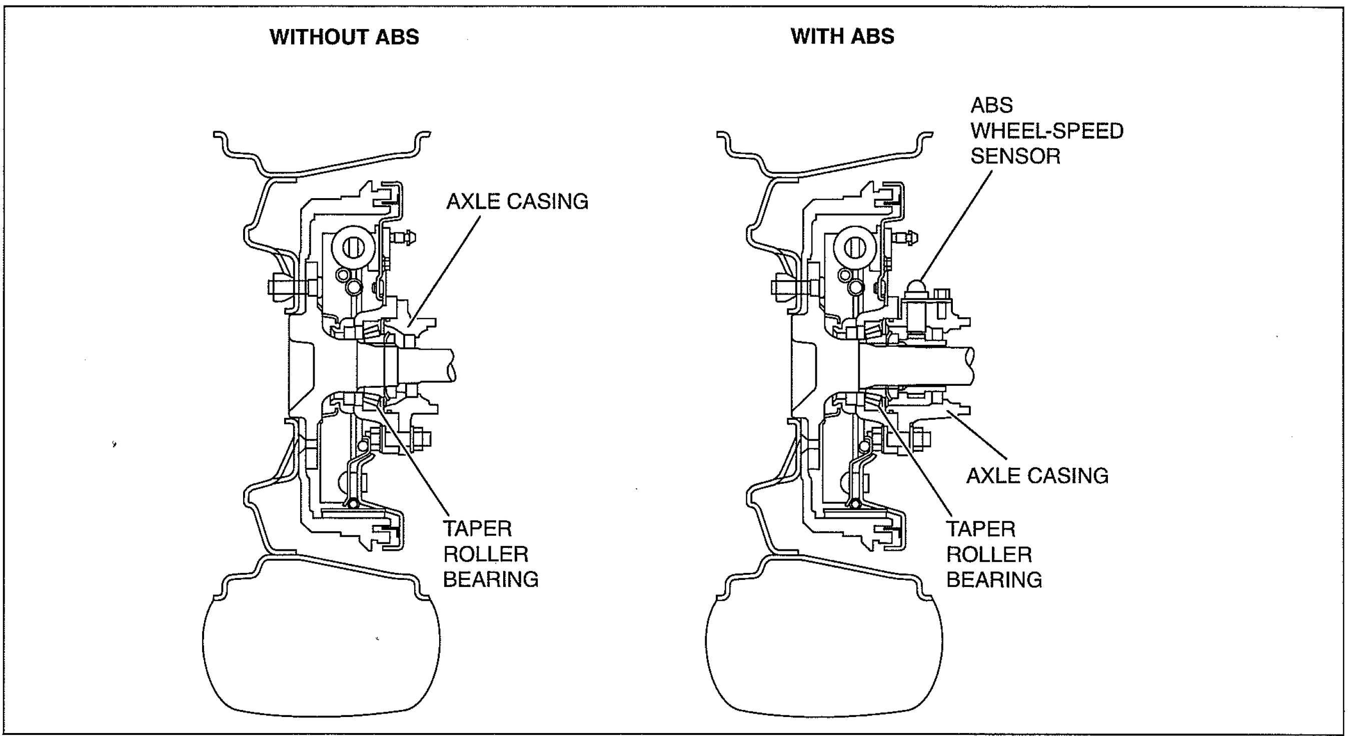
## REAR AXLE OUTLINE

dcf031205000t01

- Taper roller bearings are used as wheel bearings.
- Bearing end play in the axial direction is adjusted using shims between the rear wheel hub and the axle casing.

## REAR AXLE CROSS-SECTIONAL VIEW

dcf031205000t02




DBG312ZTB001

# 03-13 DRIVE SHAFT



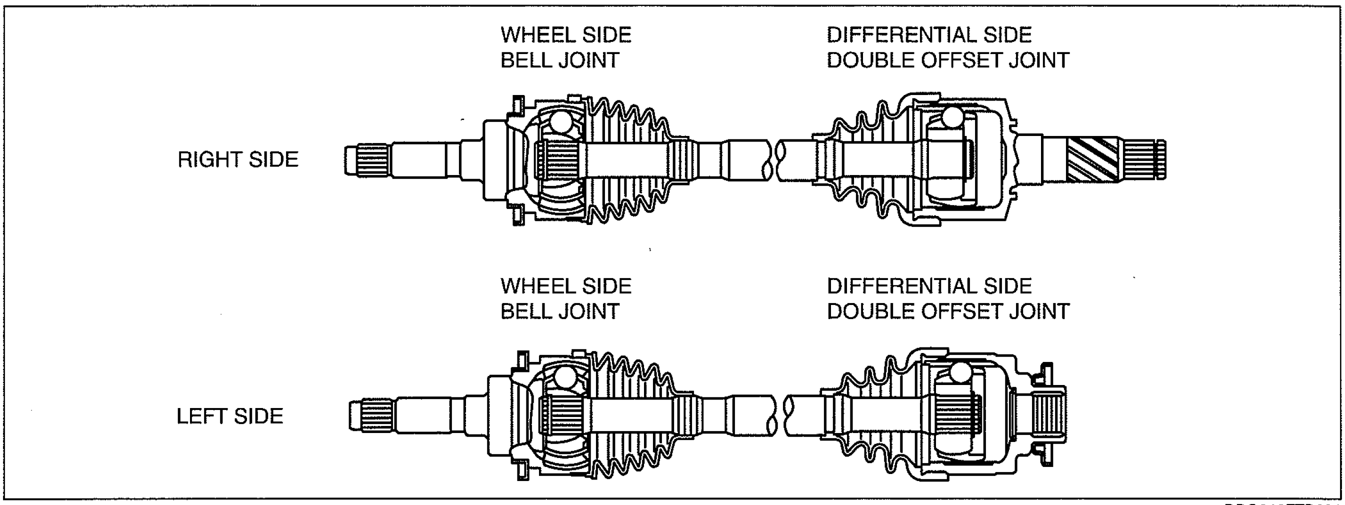
## FRONT DRIVE SHAFT OUTLINE

dcf031325500t01

- The bell joint has been adopted on the wheel side and the double offset joint has been adopted on the differential side to reduce noise and vibration.

## FRONT DRIVE SHAFT CROSS-SECTIONAL VIEW

dcf031325500t02




DBG313ZTB001

# 03-14 DIFFERENTIAL



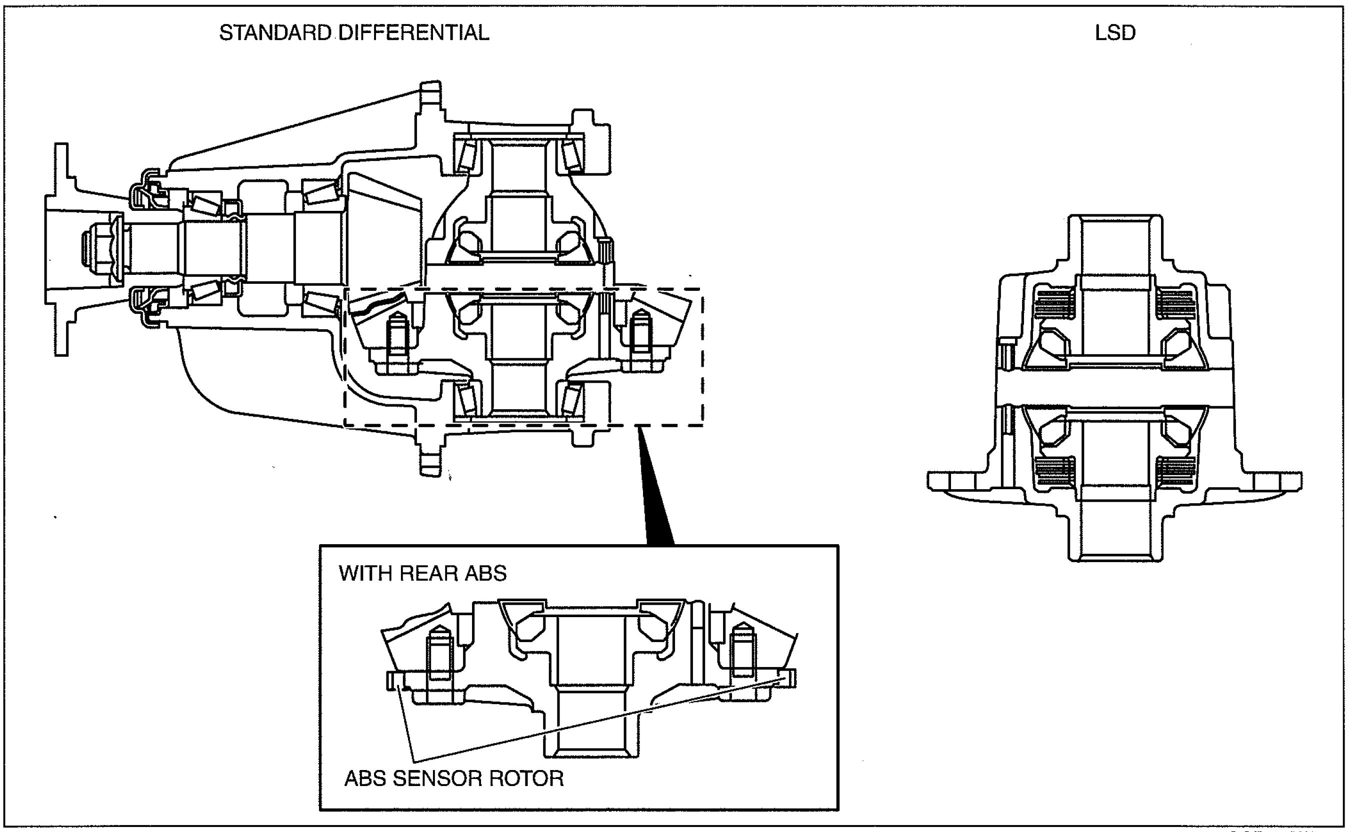
## REAR DIFFERENTIAL OUTLINE

dcf031427100t01

- There are two types of differentials available, standard differential and limited slip differential (LSD).
- The purpose of the LSD is to provide driving force to both tires by preventing either tire from slipping.
- With a standard differential, if a tire starts spinning due to poor traction such as mud or snow, the driving force is exerted on that tire by the action of the differential. With the LSD, the differential action is automatically limited and driving force is exerted on both tires.

## REAR DIFFERENTIAL CROSS-SECTIONAL VIEW

dcf031427100t02




DCF314ZTB001

# 03-16 TRANSFER [5R55S]



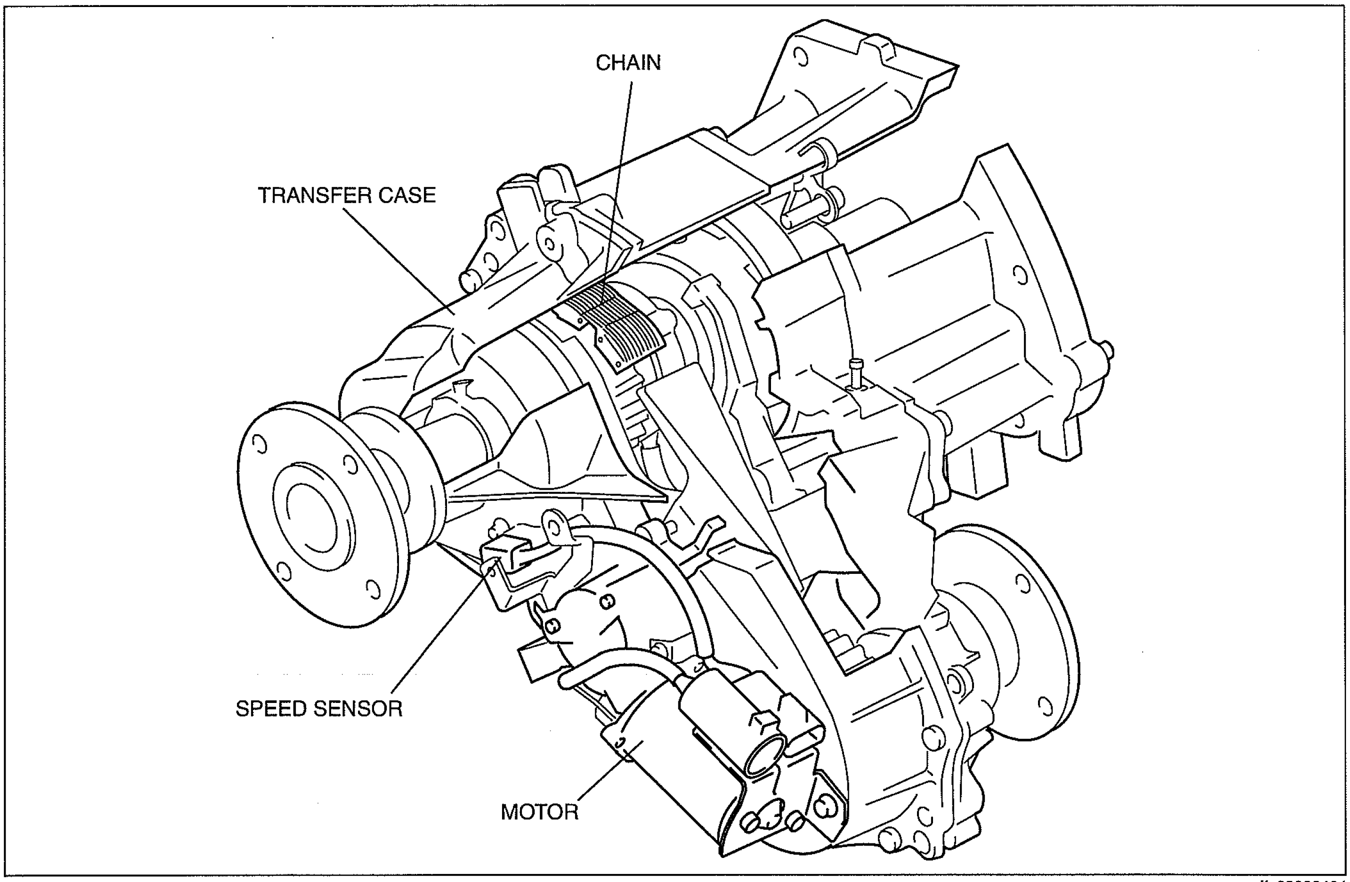
## TRANSFER CASE OUTLINE [5R55S]

id0316c1137500

- A chain-type transfer drive has been adopted for 4x4 driving to improve noise suppression. In addition, the drive sprocket, which drives the chain, rotates only during 4x4 driving to improve noise suppression during 4x2 driving.
- Shifting between 4x2 HIGH, 4x4 HIGH, and 4x4 LOW is performed by the motor for improved operability.

## TRANSFER CASE STRUCTURAL VIEW [5R55S]

id0316c1100500

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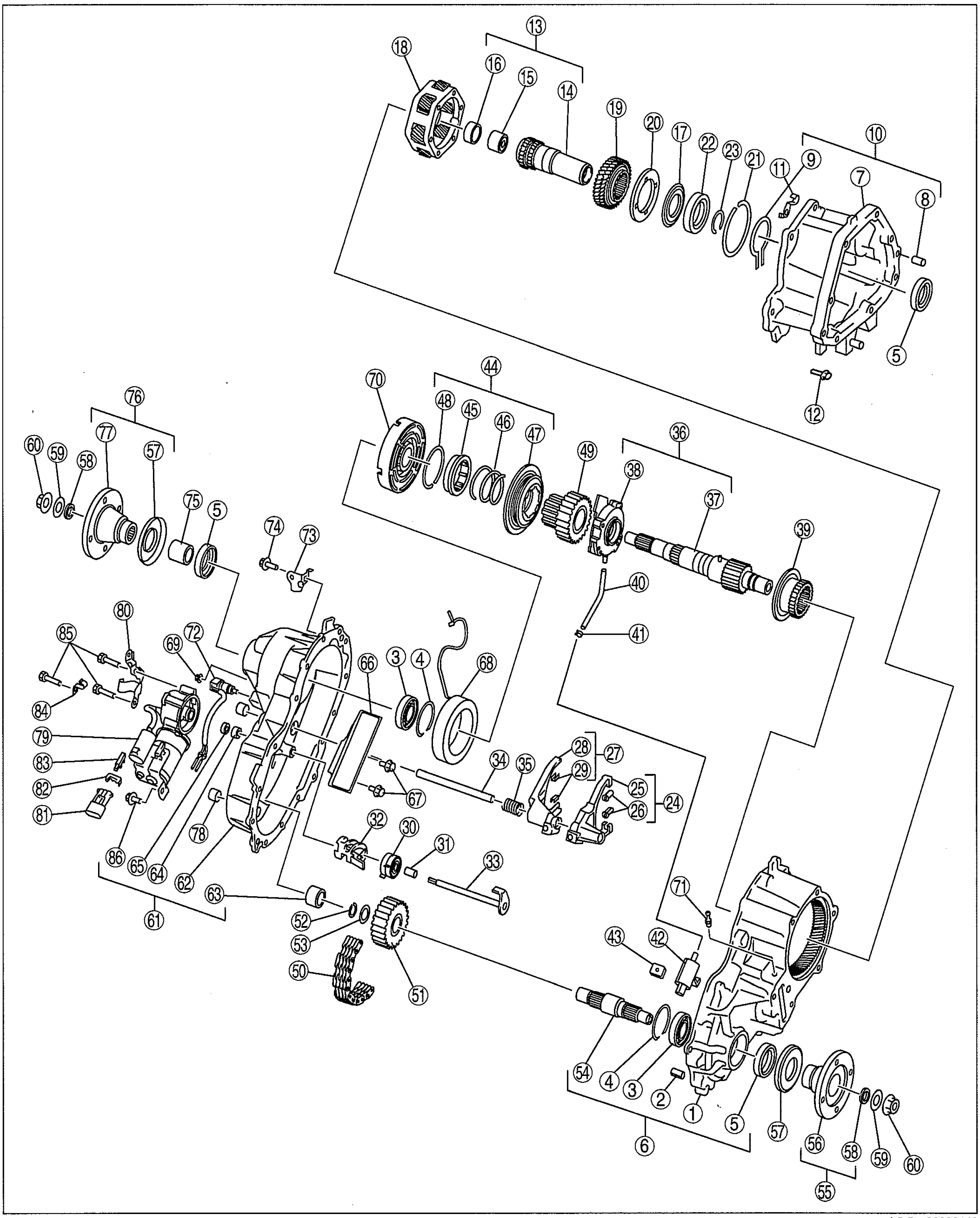
## Exploded view



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| | | | |
|---|---|---|---|
| 1 | Center transfer case | 6 | Center transfer case component |
| 2 | Dowel pin | 7 | Front transfer case |
| 3 | Bearing | 8 | Dowel pin |
| 4 | Snap ring | 9 | Snap ring |
| 5 | Oil seal | 10 | Front transfer case component |

| | |
|---|---|
| 11 | Clip |
| 12 | Bolt |
| 13 | Input shaft component |
| 14 | Input shaft |
| 15 | Needle bearing |
| 16 | Bushing |
| 17 | Thrust washer |
| 18 | Planetary gear component |
| 19 | Sun gear |
| 20 | Thrust plate |
| 21 | Snap ring |
| 22 | Ball bearing |
| 23 | Snap ring |
| 24 | Reduction shift fork component |
| 25 | Reduction shift fork |
| 26 | Reduction shift fork facing |
| 27 | Lockup shift fork component |
| 28 | Lockup shift fork |
| 29 | Lockup shift fork facing |
| 30 | Spring |
| 31 | Spacer |
| 32 | Cam |
| 33 | Shift shaft |
| 34 | Rail shift |
| 35 | Return spring |
| 36 | Output shaft and gerotor pump component |
| 37 | Output shaft component |
| 38 | Gerotor pump component |
| 39 | Reduction hub |
| 40 | Pump hose |
| 41 | Hose clamp |
| 42 | Oil strainer |
| 43 | Magnet |
| 44 | Lockup component |
| 45 | Lockup hub |
| 46 | Spring |
| 47 | Lockup collar |
| 48 | Snap ring |

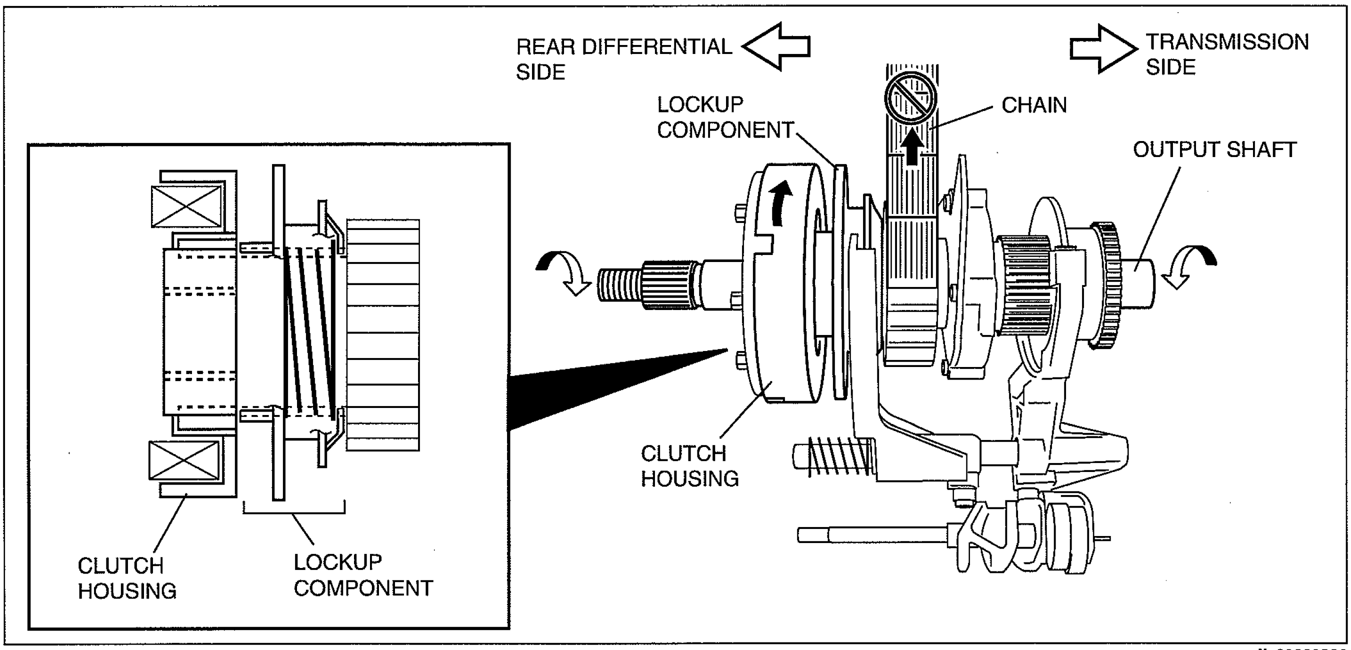
| | |
|---|---|
| 49 | Drive sprocket |
| 50 | Drive chain |
| 51 | Driven sprocket |
| 52 | Snap ring |
| 53 | Spacer |
| 54 | Lower output shaft |
| 55 | Front output flange component |
| 56 | Front output flange |
| 57 | Deflector |
| 58 | Oil seal |
| 59 | Washer |
| 60 | Locknut |
| 61 | Rear transfer case component |
| 62 | Rear transfer case |
| 63 | Needle bearing |
| 64 | Sleeve |
| 65 | Seal |
| 66 | Snubber |
| 67 | Bolt |
| 68 | Clutch coil |
| 69 | Nut |
| 70 | Clutch housing |
| 71 | Breather pipe |
| 72 | Speed sensor component |
| 73 | Bracket |
| 74 | Bolt |
| 75 | Spacer |
| 76 | Rear companion flange component |
| 77 | Rear companion flange |
| 78 | Oil plug |
| 79 | Motor component |
| 80 | Bracket |
| 81 | Terminal connector |
| 82 | Clip |
| 83 | Bracket |
| 84 | Clip |
| 85 | Bolt |
| 86 | Bolt |

## TRANSFER CASE POWER FLOW [5R55S]

id0316c1100600

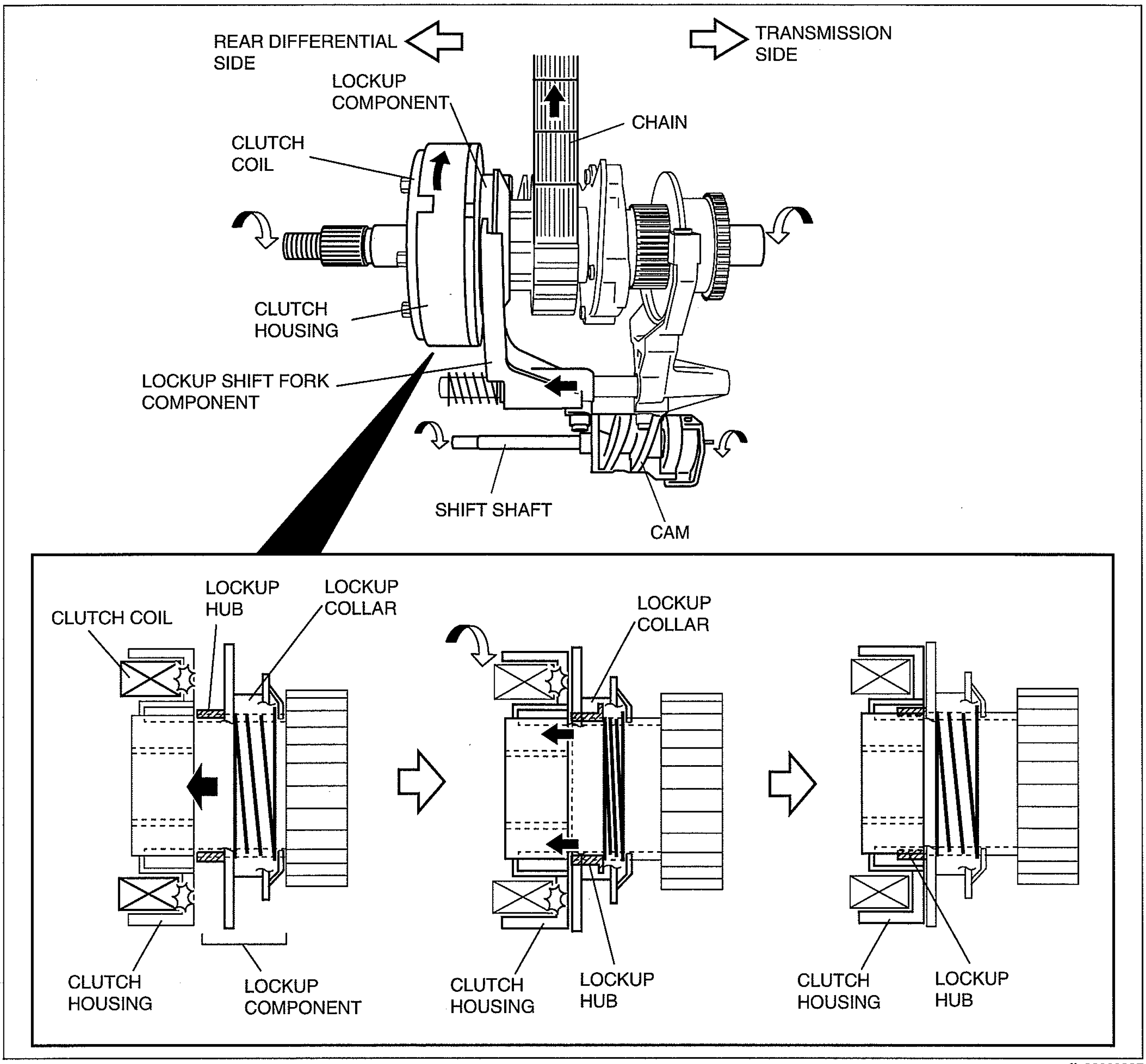
### Shifting From 4x2 HIGH to 4x4 HIGH

- During 4x2 driving, drive force from the transmission is transmitted from the input shaft to the output shaft through the reduction hub, and the drive force from the output shaft is transmitted to the rear differential Shifting between 4x2 and 4x4 is performed using the lockup component operation. During 4x2 driving, drive force is not transmitted to the chain because the lockup component does not contact the clutch housing, therefore, the drive force is not transmitted to the front differential.




arnffn00000336

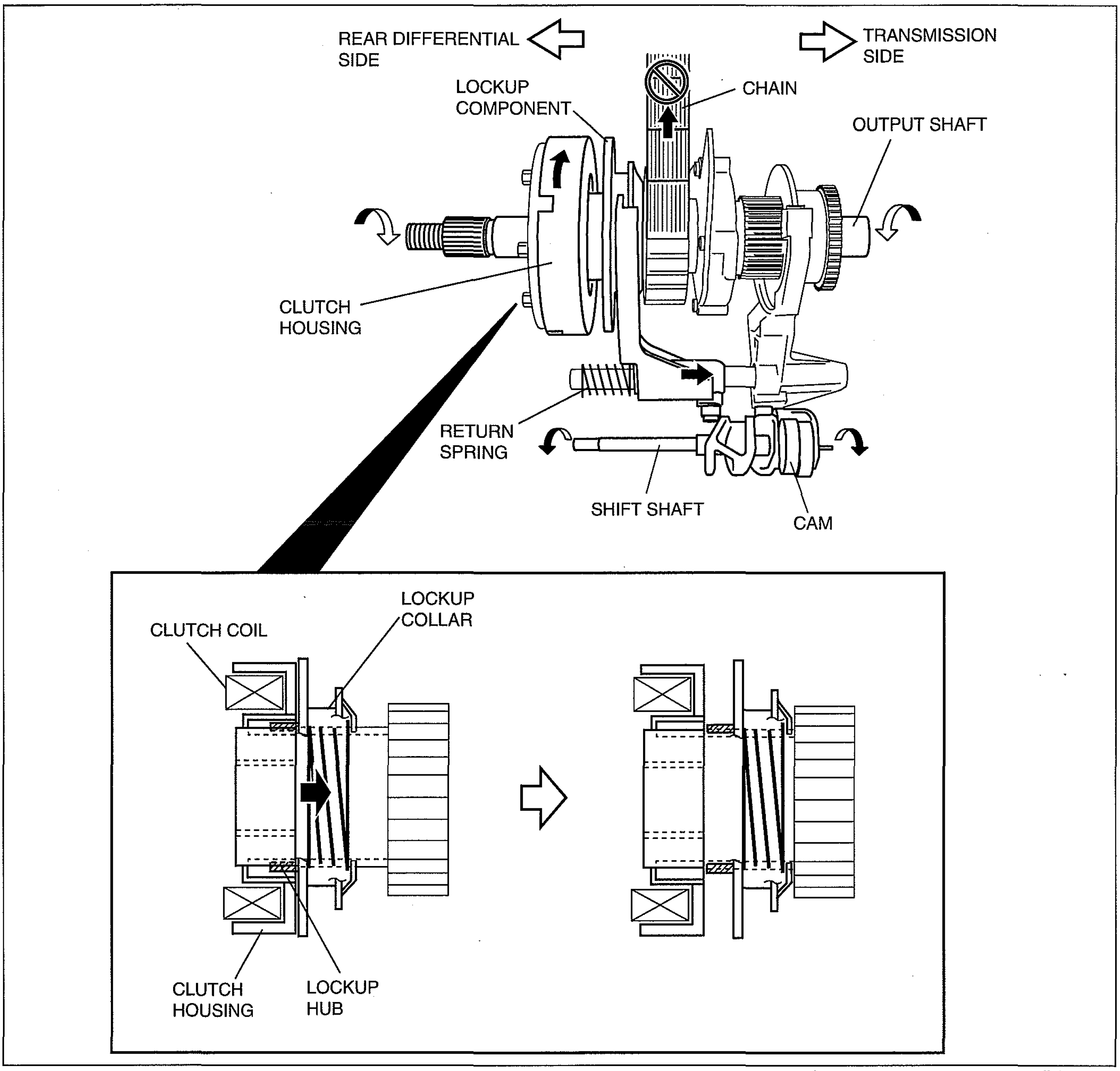
- When the 4x4 switch is switched to the 4H position during 4x2 driving, the motor rotates from the 2H to 4H position. Due to the motor rotation, the cam rotates via the shift shaft. When the cam rotates, the lockup shift fork component slides to the clutch housing side. In conjunction with this operation, the clutch coil is energized and magnetic force is generated, and the lockup shift fork component is pulled toward the clutch housing. In the lockup shift fork component operation, the lockup hub contacts the clutch housing first, then the lockup collar slides to the lockup housing side. At this time, if the lockup hub is not engaged with the clutch housing spline, the lock up hub is pressed into the lockup collar. If the clutch housing rotates in this condition, such as when the vehicle is being driven, the lockup hub engages with the clutch housing spline. The lockup shift fork component is engaged with the drive sprocket and the clutch housing is engaged with the output shaft, therefore, drive force from the output shaft is transmitted to the lower output shaft via the chain. Due to this, drive force is transmitted to the front differential.




arnffn00000337

### Shifting From 4x4 HIGH to 4x2 HIGH

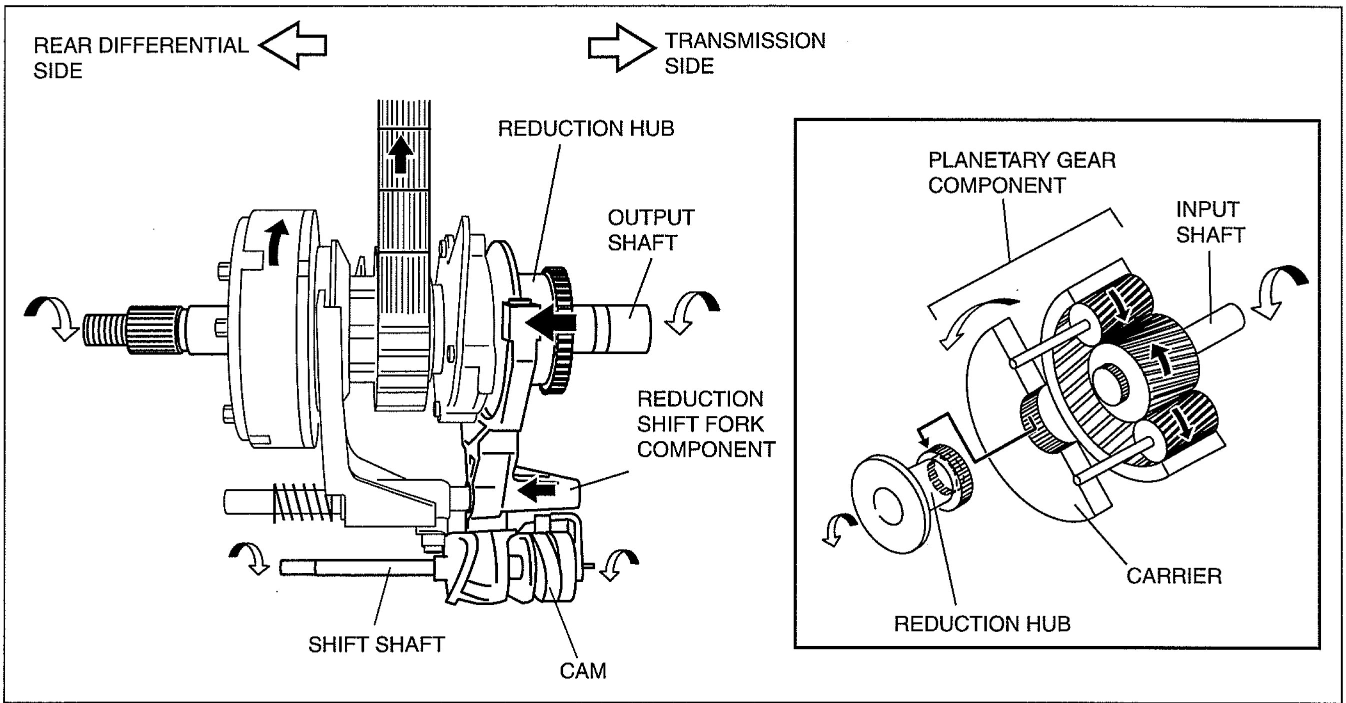
- When the 4x4 switch is switched to the 2H position while in 4x4 HIGH, the motor rotates from the 4H to 2H position. Due to the motor rotation, the cam rotates via the shift shaft. The lockup shift fork component slides due to the cam rotation and spring force of the return spring, and separates from the clutch housing. At this time, the clutch coil is not energized.When the lockup shift fork component separates from the clutch housing, drive force transmission to the lower output side is interrupted, and drive force is transmitted only to the rear differential.




amffn00000338

### Shifting From 4x4 HIGH to 4x4 LOW

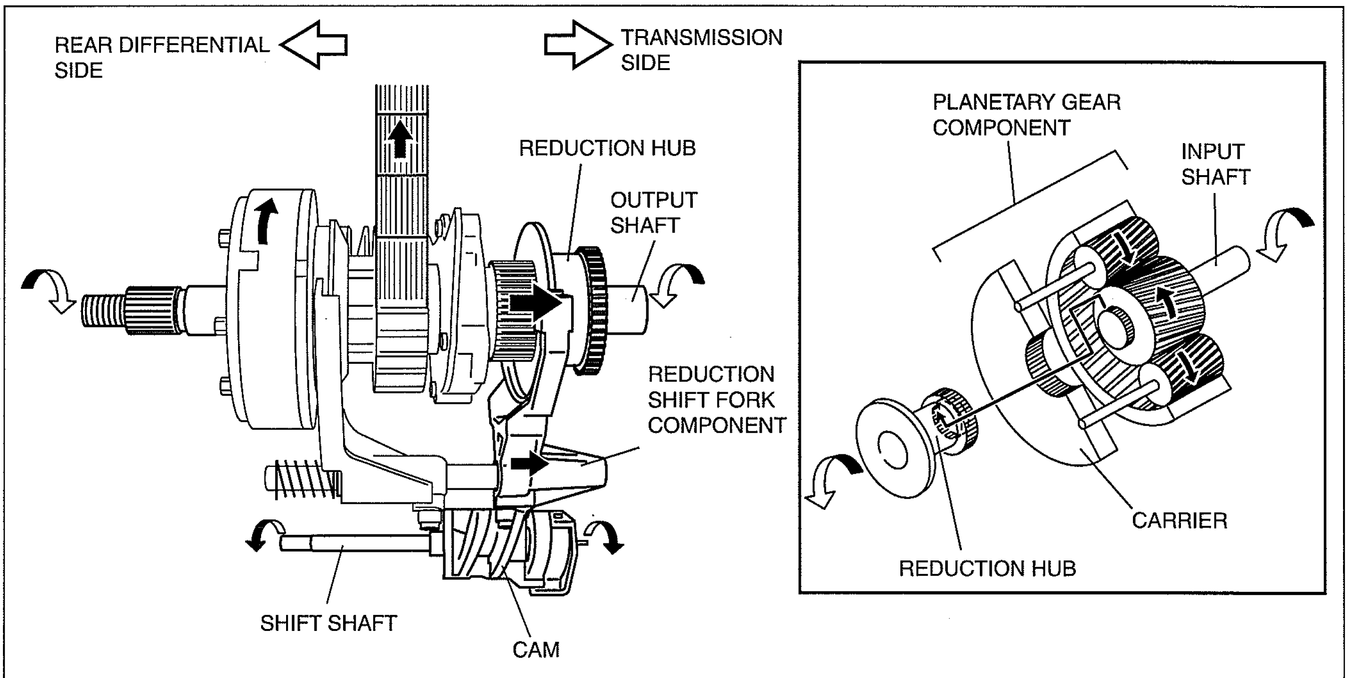
- When the 4x4 switch is switched to the 4L position while in 4x4 HIGH, the motor rotates from the 4H to 4L position. Due to the motor rotation, the cam rotates via the shift shaft. When the cam rotates the reduction shift fork component, which is fitted into the cam groove, slides, and the reduction hub engages with the spline which is inside the carrier of the planetary gear component. If the drive force is transmitted to the input shaft in this condition, the output speed is reduced due to the planetary gear operation, and the rotational torque increases.This drive force is transmitted to the front and rear differentials.




arnffn00000339

### Shifting From 4x4 LOW to 4x4 HIGH

- When the 4x4 switch is switched to the 4H position while in 4x4 LOW, the motor rotates from the 4L to 4H position. Due to the motor rotation, the cam rotates via the shift shaft. When the cam rotates the reduction shift fork component, which is fitted into the cam groove, slides, and the reduction hub engages with the input shaft spline. Due to this, the drive force from the input shaft is transmitted to the output shaft without being input to the planetary gear, and then it is transmitted to the front and rear differential with the output speed and rotational torque remaining unchanged.




arnffn00000340

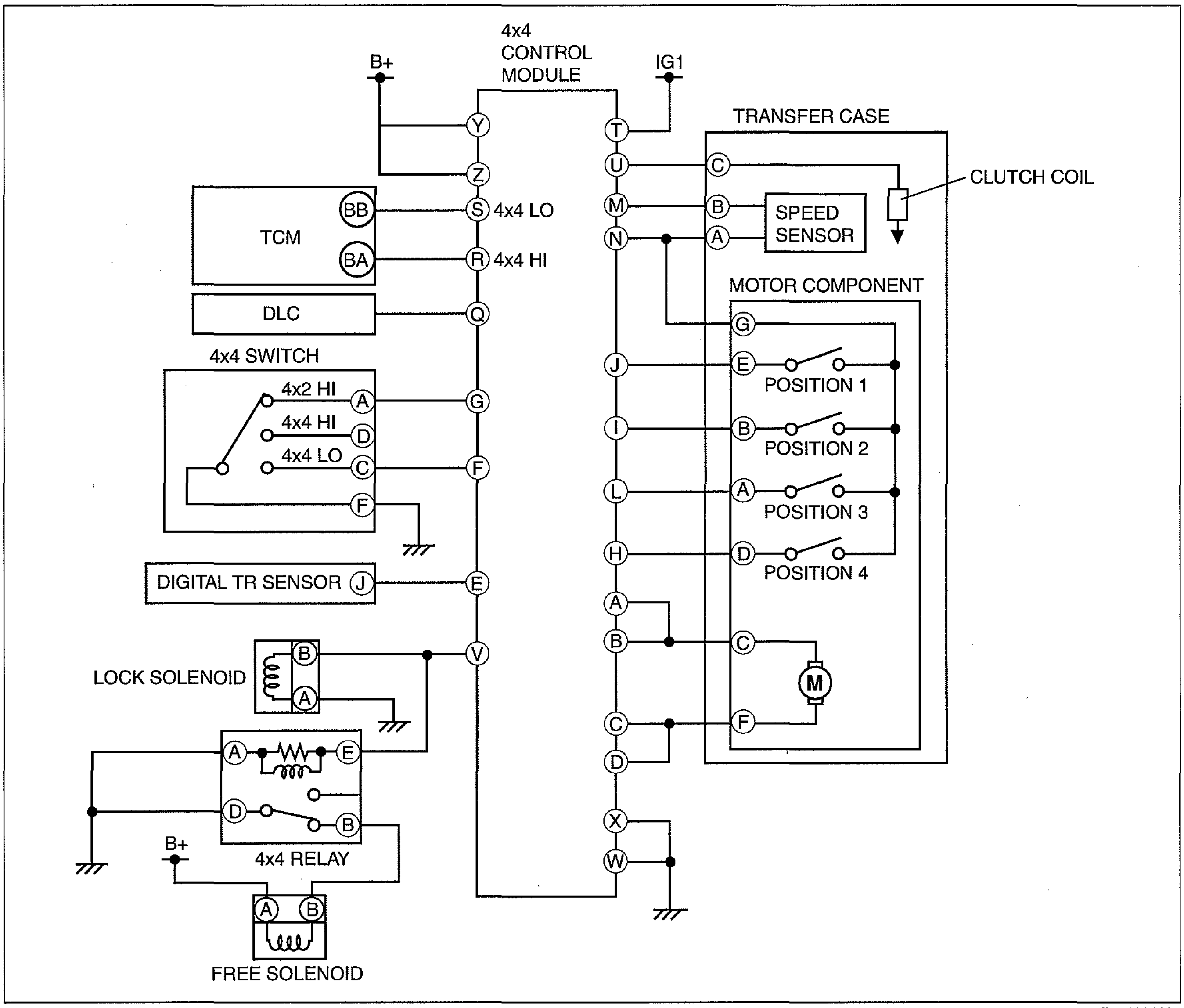
## TRANSFER CASE ELECTRONIC CONTROL SYSTEM OUTLINE [5R55S]

id0316c1137600

- A 4x4 control module has been adopted which operates the motor according to signals from the speed sensor and motor position detection switch to control shifting between 4x2 HIGH, 4x4 HIGH, and 4x4 LOW.

## TRANSFER CASE SYSTEM WIRING DIAGRAM [5R55S]

id0316c1137700

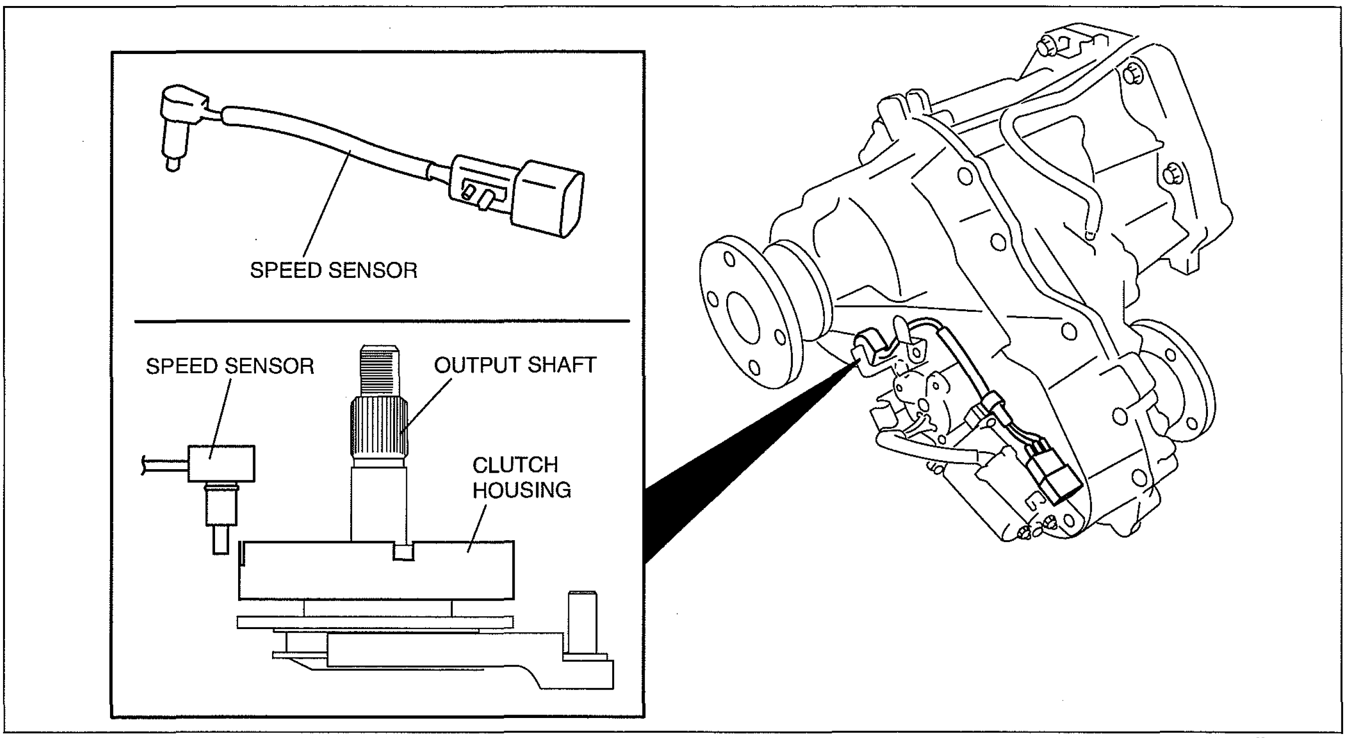



arnffw00001937

## SPEED SENSOR CONSTRUCTION [5R55S]

id0316c1137000

- The speed sensor detects the output shaft rotation speed, and the signal is input to the 4x4 control module.
- Signals from the speed sensor are used by the 4x4 control module to perform each control.




arnffn00000341

## MOTOR POSITION DETECTION SWITCH FUNCTION [5R55S]

id0316c1137100

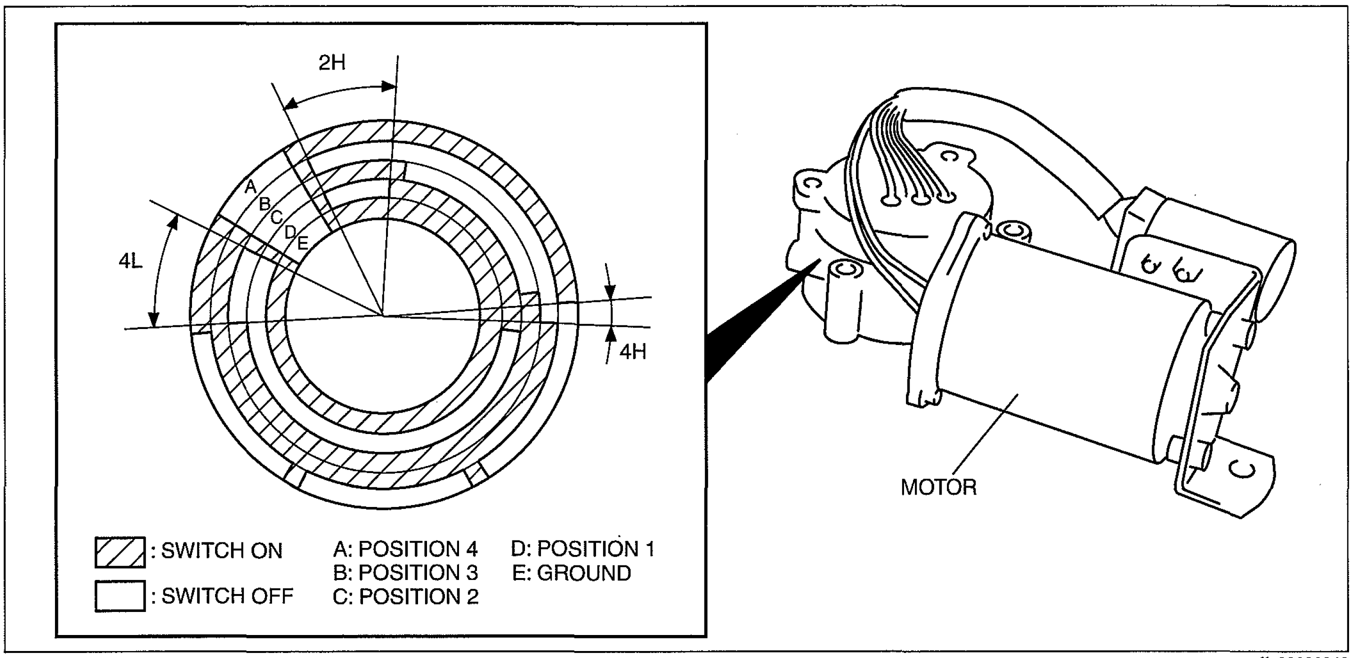
- The motor position detection switch detects the motor position and outputs control signals to the 4x4 control module.
- The 4x4 control module determines the motor position according to the signals from the motor position detection switch.

## MOTOR POSITION DETECTION SWITCH CONSTRUCTION/OPERATION [5R55S]

id0316c1137200

- The motor position detection switch is integrated into the motor, and turns on or off according to the motor position.
- The motor position detection switch detects the motor position using a combination of four switches.

| Motor Position | Position 1 | Position 2 | Position 3 | Position 4 |
|---|---|---|---|---|
| 2H | Off | On | Off | On |
| 4H | On | On | Off | Off |
| 4L | Off | Off | On | On |

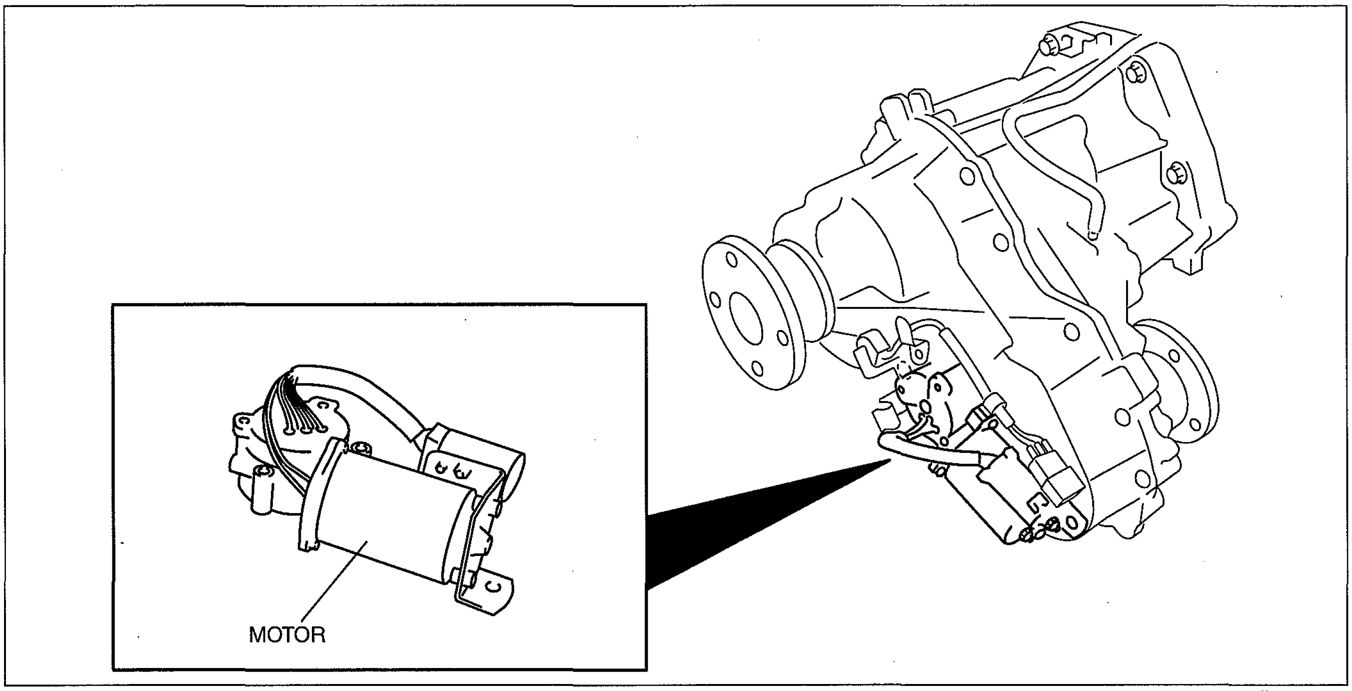



arnffn00000342

## MOTOR CONSTRUCTION [5R55S]

id0316c1137300

- The motor switches between 4x2 HIGH, 4x4 HIGH, and 4x4 LOW according to the signals from the 4x4 control module.
- A switch for motor position detection is integrated in the motor.

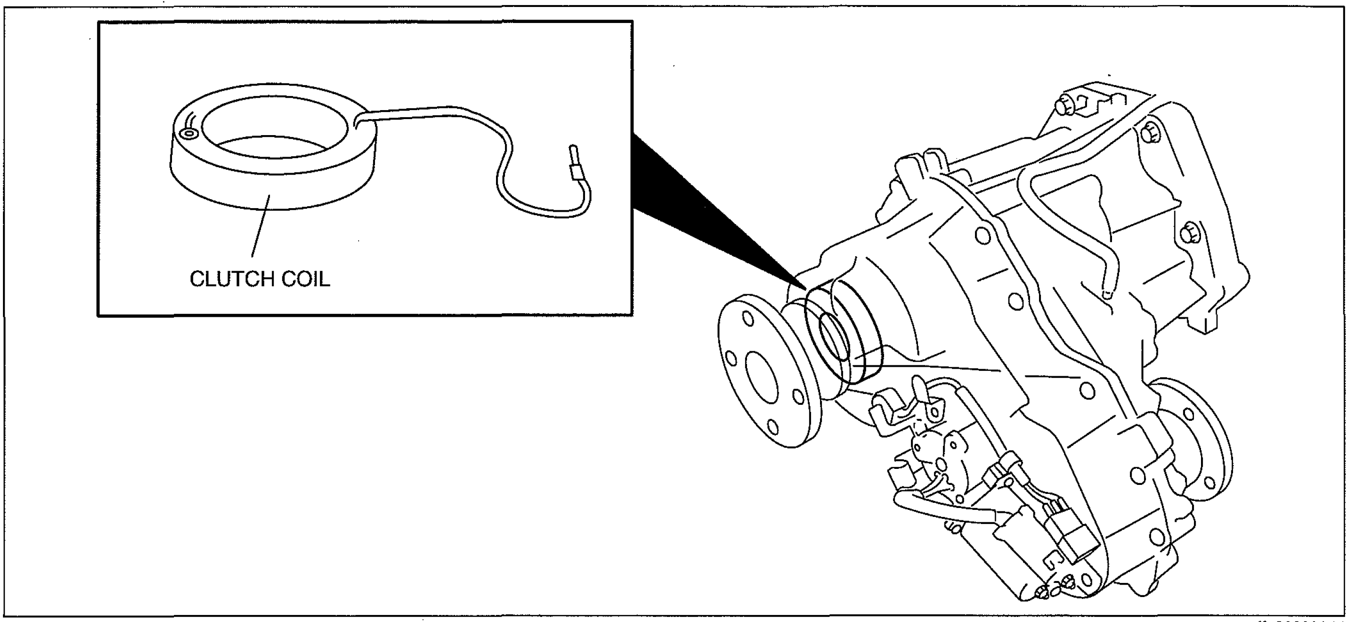



arnffn00000343

## CLUTCH COIL CONSTRUCTION [5R55S]

id0316c1137400

- The clutch coil is integrated in the transfer.
- When a 4x4 shift signal is input to the 4x4 control module, electricity flows from the control module to the clutch coil and magnetic force is generated. Using the magnetic force, the lockup collar is pulled toward the clutch housing, shifting from 4x2 to 4x4.




arnffn00000344

# 03-16 TRANSFER



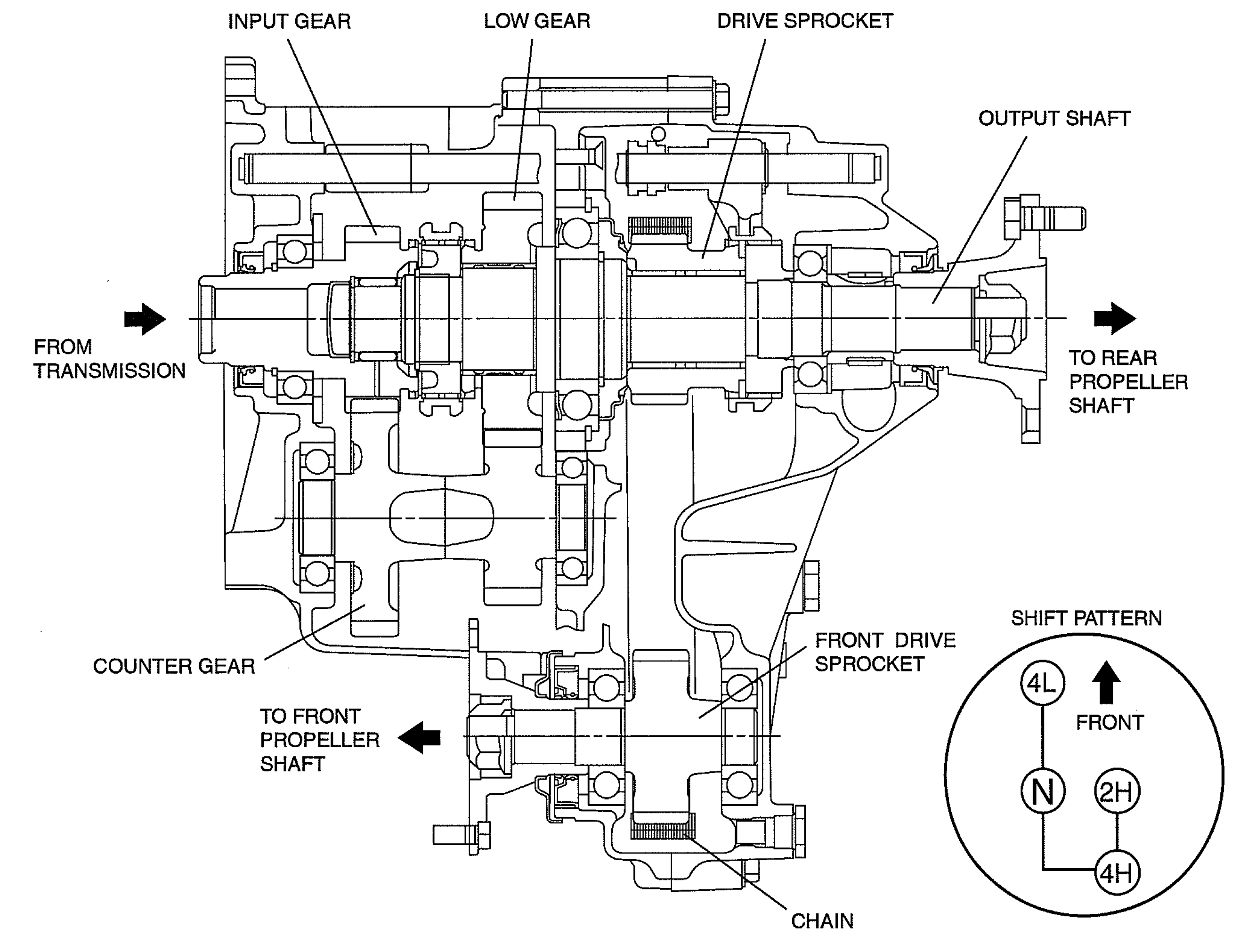
## TRANSFER OUTLINE

dcf031600000t01

- The transfer of S15MX-D manual transmission is a chain-drive part-time transfer in which the shift activates mechanically.
- The speed selection is controlled by a single-lever shift mechanism that provides N, 2H, 4H and 4L.
- The purpose of the transfer is for transmission of driving force to either the rear differential only or to the front and rear differentials simultaneously.
- The transfer employs a chain to transfer the driving force to the front differential.
- The transfer is a non-synchro type transmission.
- The chain is a maintenance-free type, requiring no tension adjustment.

## TRANSFER CROSS-SECTIONAL VIEW

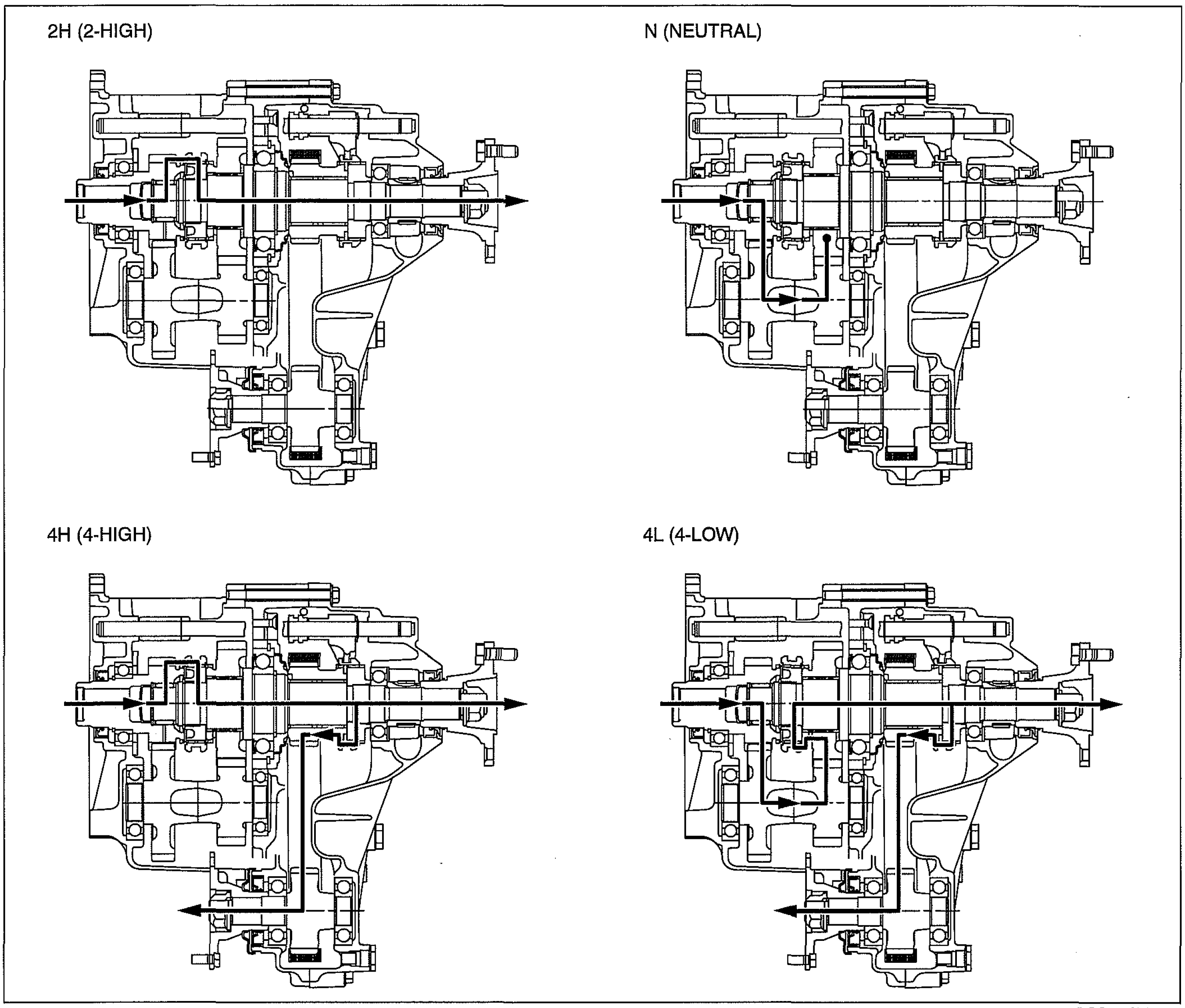
dcf031600000t02




DCG316ZTB001

## TRANSFER POWER FLOW

dcf031600000t03




DCG316ZTB002

# 03-18 4-WHEEL DRIVE



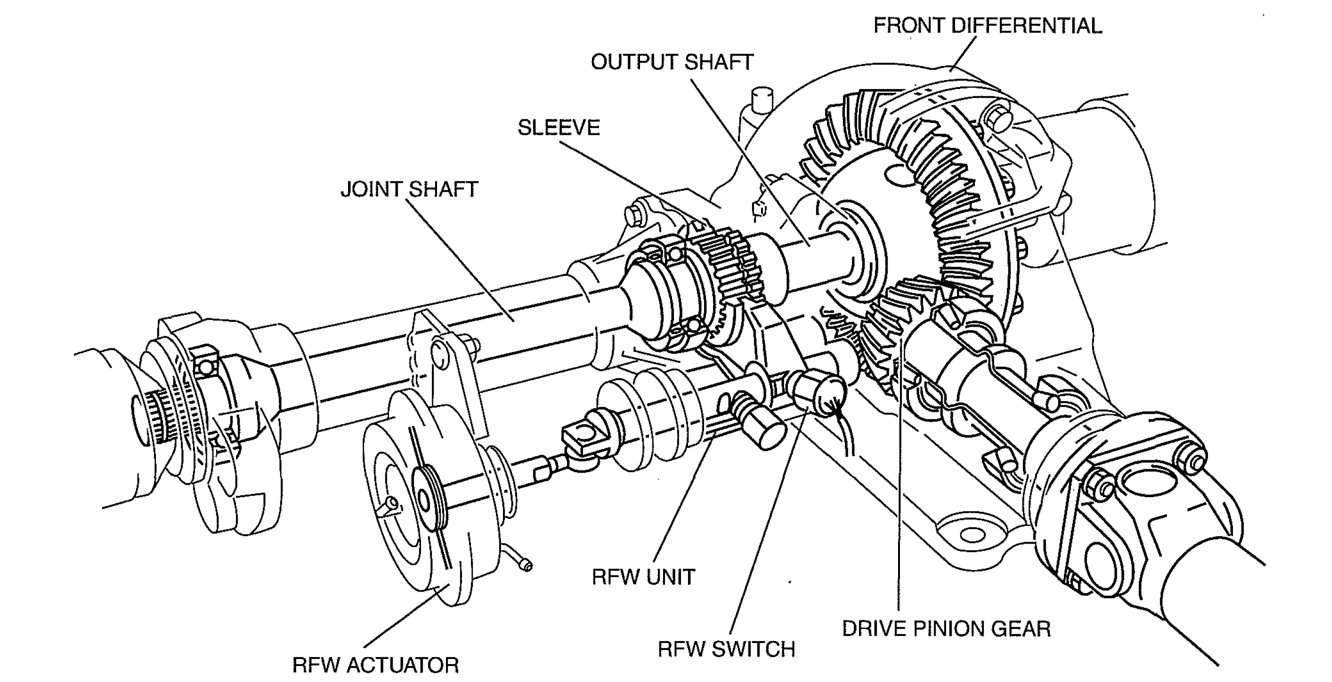
## REMOTE FREEWHEEL (RFW) OUTLINE

dcf031827300t01

- The PCM receives signals from the transfer case switch and other switches. It then activates one of two solenoids, causing a vacuum from the vacuum pump to be applied to the RFW actuator, and switching the remote freewheel (RFW) unit to lock or free.
- To prevent reduced performance of the actuator as a result of a decrease in engine vacuum or vacuum pump, such as during acceleration or at high altitude operation, the actuation system includes a one-way check valve.
- The switching mechanism of the RFW unit is a mechanical dog clutch on the left side of the front differential.
- The RFW operation is controlled by PCM.

## REMOTE FREEWHEEL (RFW) CONSTRUCTION

dcf031827300t02




DBG314ZTB012

03

## REMOTE FREEWHEEL (RFW) COMPONENTS AND FUNCTIONS

dcf031827300t03

### Electrical component

| Part name | | | Function |
|---|---|---|---|
| PCM (RFW control) | | | • Sends ON/OFF signals to lock and free solenoids and indicator lights according to signals from various switches |
| Input | Transfer neutral switch | | • Detects transfer select lever N position |
| | 4x4 indicator light | | • Detects transfer select lever 2H position |
| | RFW switch | | • Detects RFW unit "lock" |
| | RFW main switch | | • Creates RFW unit "free" |
| Output | Solenoid | | • Switched ON/OFF by electrical signals from PCM<br>• Regulates RFW unit "lock" or "free" |
| | | Lock | • Switched ON when "locked" |
| | | Free | • Switched ON when "free" |
| | 4x4 indicator light | | • Illuminates when 4H or 4L selected |
| | RFW indicator light | | • Illuminates when RFW unit locked |

### Mechanical component

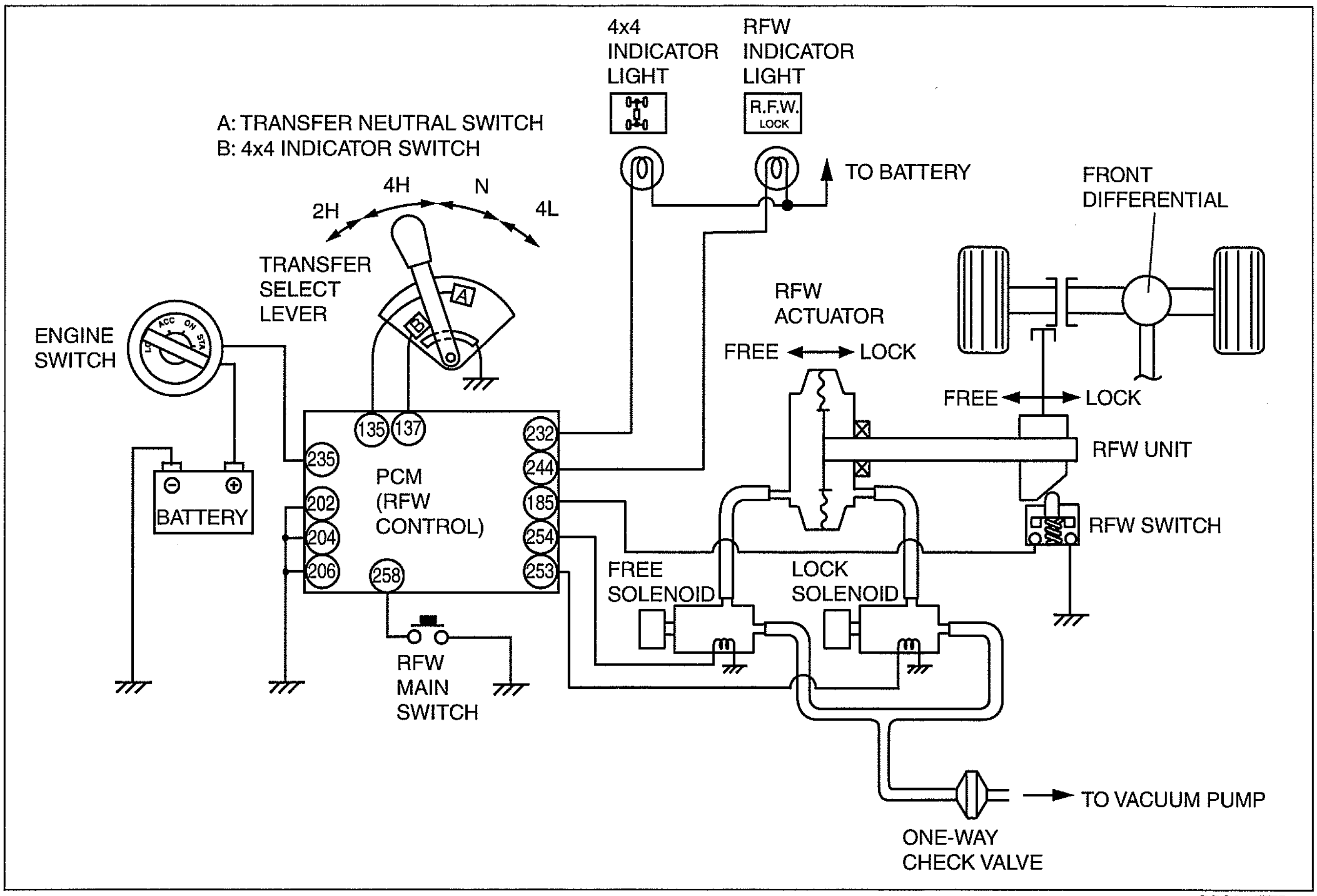
| Part name | Function |
|---|---|
| Transfer select lever | • Sets transfer case operation mode (2H, 4H, N, or 4L) |
| RFW unit | • Transmits front propeller shaft rotation to front tires |
| RFW actuator | • "Locks" or "free" RFW unit |
| One-way check valve | • Prevents leakage of vacuum |

### Relationship of Component and Function

| Part name / Function | Input | | | | Output | | | |
|---|---|---|---|---|---|---|---|---|
| | Transfer neutral switch | 4x4 indicator switch | RFW switch | RFW main switch | Lock solenoid | Free solenoid | 4x4 indicator light | RFW indicator light |
| Controlled by driver | x | x | | x | | | | |
| Controlled by RFW unit | | | x | | | | | |
| Controlled by PCM | | | | | x | x | x | x |

## REMOTE FREEWHEEL (RFW) SYSTEM DIAGRAM

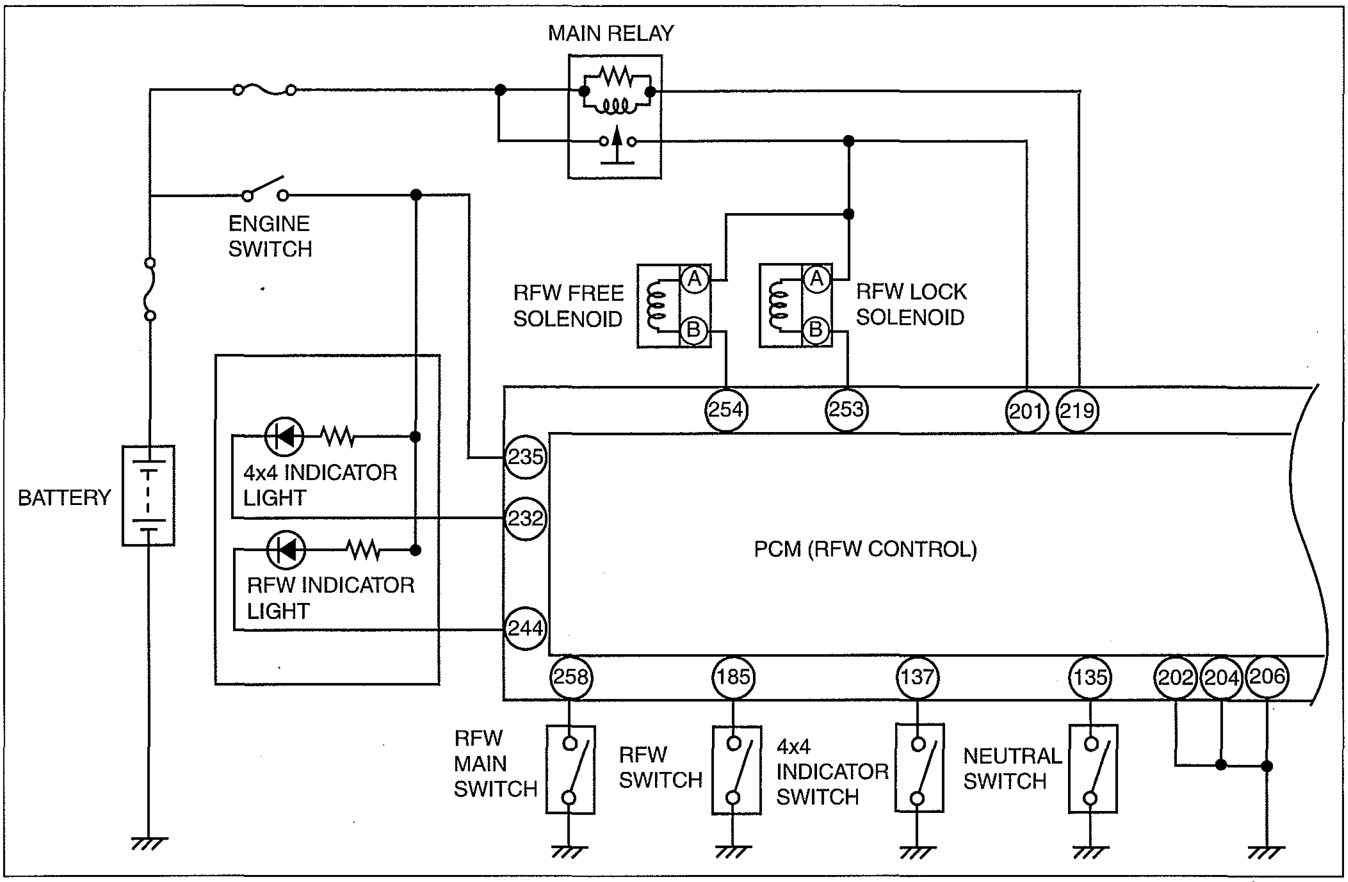
dcf031827300t04




DBG314ZTB003

## REMOTE FREEWHEEL (RFW) SYSTEM WIRING DIAGRAM

dcf031827300t05




DBG318ZTB001

## REMOTE FREEWHEEL (RFW) OPERATION

dcf031827300t06

### 2H (free) to 4H Selection

- The remote freewheel (RFW) unit is automatically locked when the select lever is moved from 2H to 4H while the vehicle is stopped. At this time, the 4x4 indicator light and the RFW indicator light illuminate to inform the drive that the vehicle is in four-wheel-drive mode and that the RFW unit is locked.

**Caution**
- **The changeover from 2H (free) to 4H (locked) MUST NOT be made while the vehicle is moving because it may result in damage to the front differential and RFW components.**
  **If such a shift is made by mistake, a ratching noise may be heard. In this case return the select lever to 2H and press the RFW main switch.**

### 4H to 2H (free) Selection

- The RFW unit will be automatically freed when the RFW main switch is pressed once after the select lever is moved (while the vehicle is moving or while the vehicle is stopped) from 4H to 2H. At this time, the 4x4 indicator light and the RFW indicator light go OFF to inform the driver that the vehicle is in 4x2 mode and that the RFW unit has been freed.

**Note**
- The RFW unit will remain locked only if the RFW main switch is pressed while the vehicle is operating in 4H or 4L.

### 2H (lock) to 4H, or 4H to 2H (lock) Selection

- If the RFW unit is locked, the transfer case can be changed from 2H to 4H or vice versa while the vehicle is running. At this time, the RFW indicator light remains illuminated to inform the driver that the RFW unit remains locked.

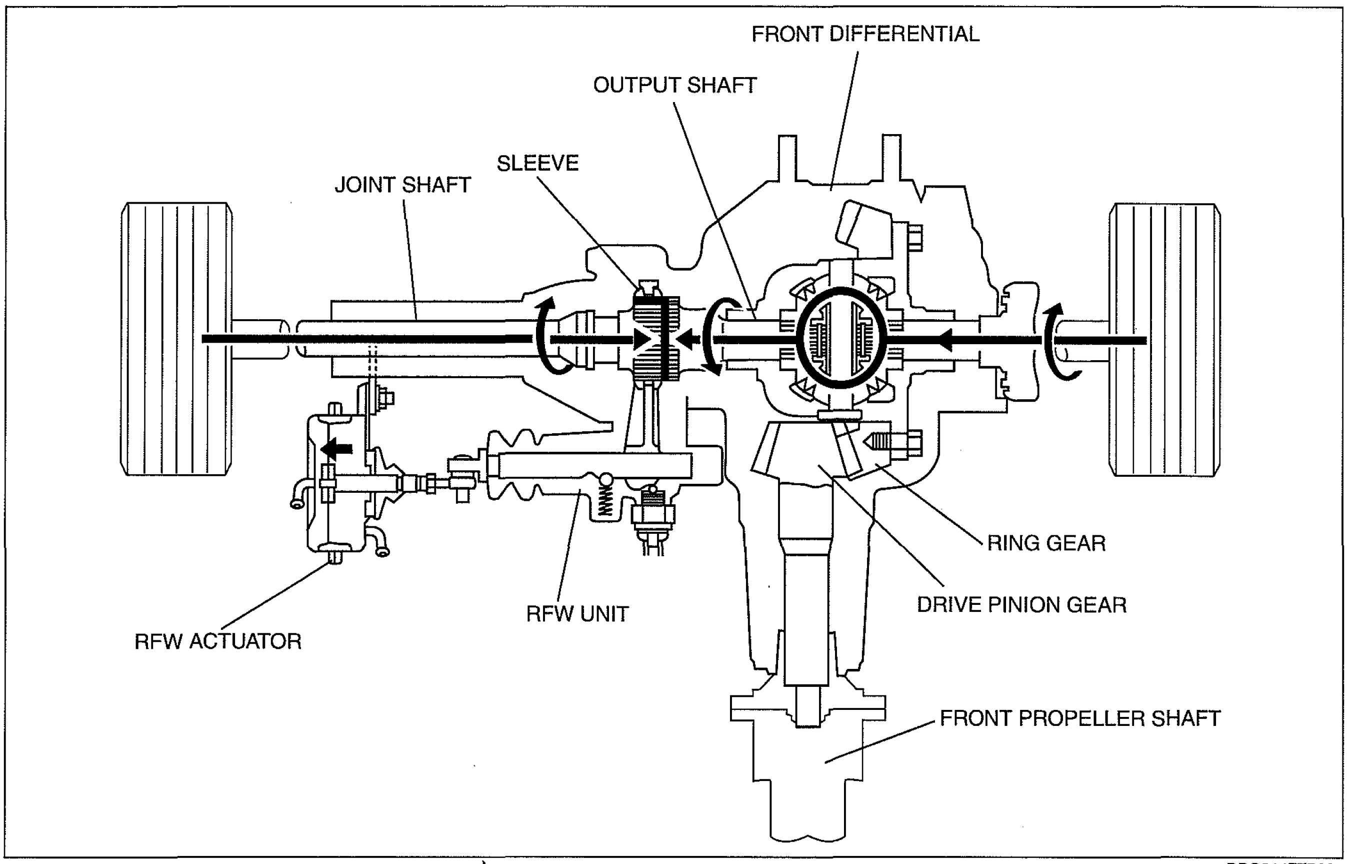
### 4H to 4L, or 4L to 4H Selection

- As with a conventional 4x4, the transfer case can be shifted from 4H to 4L or vice versa when depressing the clutch while the vehicle is stopped and the engine is running. At this time, the RFW unit remains locked, and the 4x4 and RFW indicator lights remain illuminated.

**Transmission Oil Temperature Increase Prevention**

- If the vehicle is driven at high speed with the RFW locked, the temperature of the oil for operating the transmission, transfer, and front differential increases, resulting in internal part damage. Due to this, the RFW indicator light flashes (**0.3 s** intervals) to alert the driver when the following condition is met:
  — Vehicle is driven at **100 km/h {62 MPH} or more** with the RFW locked.
  — When the vehicle speed decreases to less than **95 km/h {59 MPH}** from the above condition, the RFW indicator light illuminates again. (Normal indication)

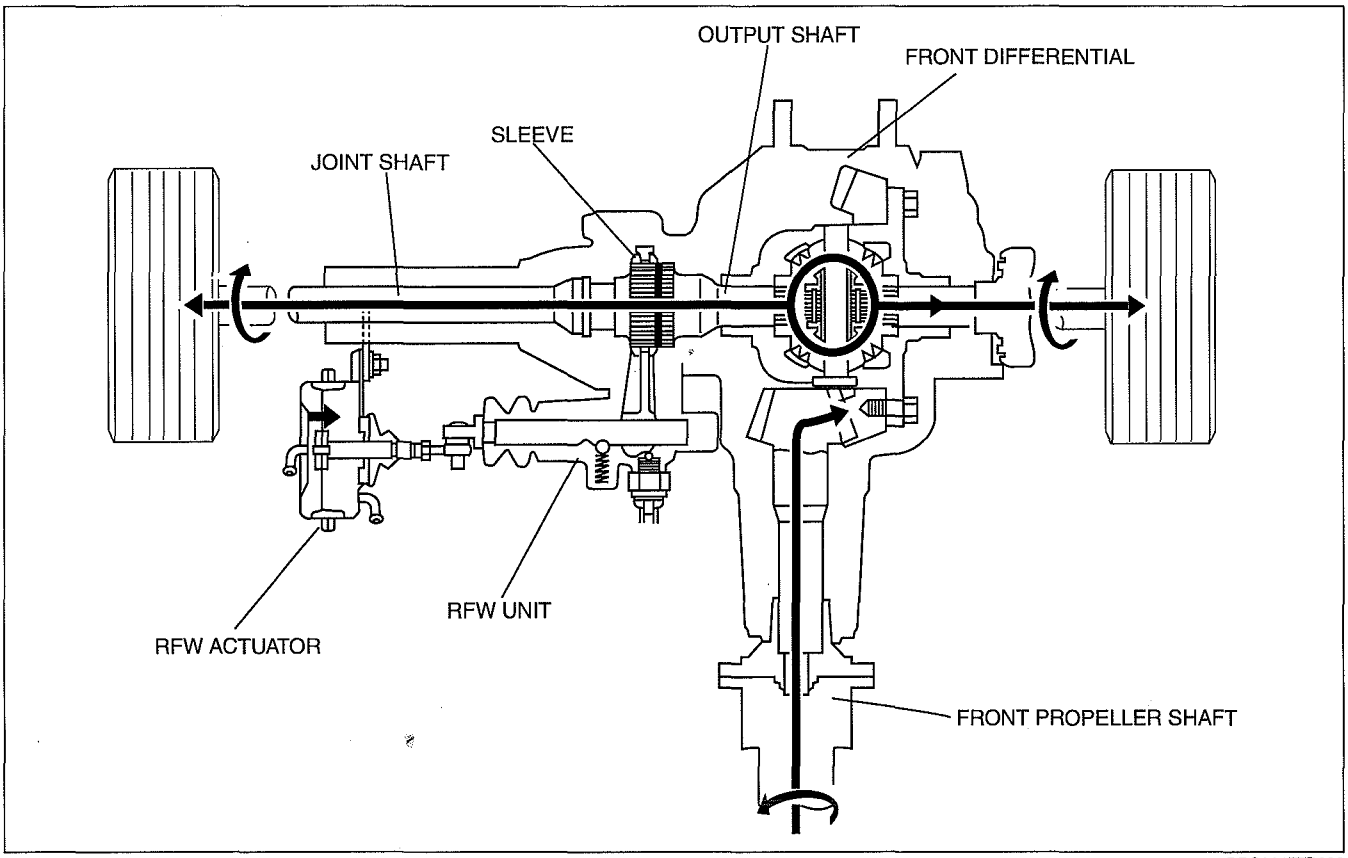
**Remote Freewheel Unit: FREE**




DBG314ZTB005

- The RFW actuator pulls the sleeve away from the front differential output shaft, allowing the front differential to rotate freely.
  The rotation of the right front tire is absorbed by the front differential, with the result that the ring gear, drive pinion gear, and front propeller shaft do not turn.

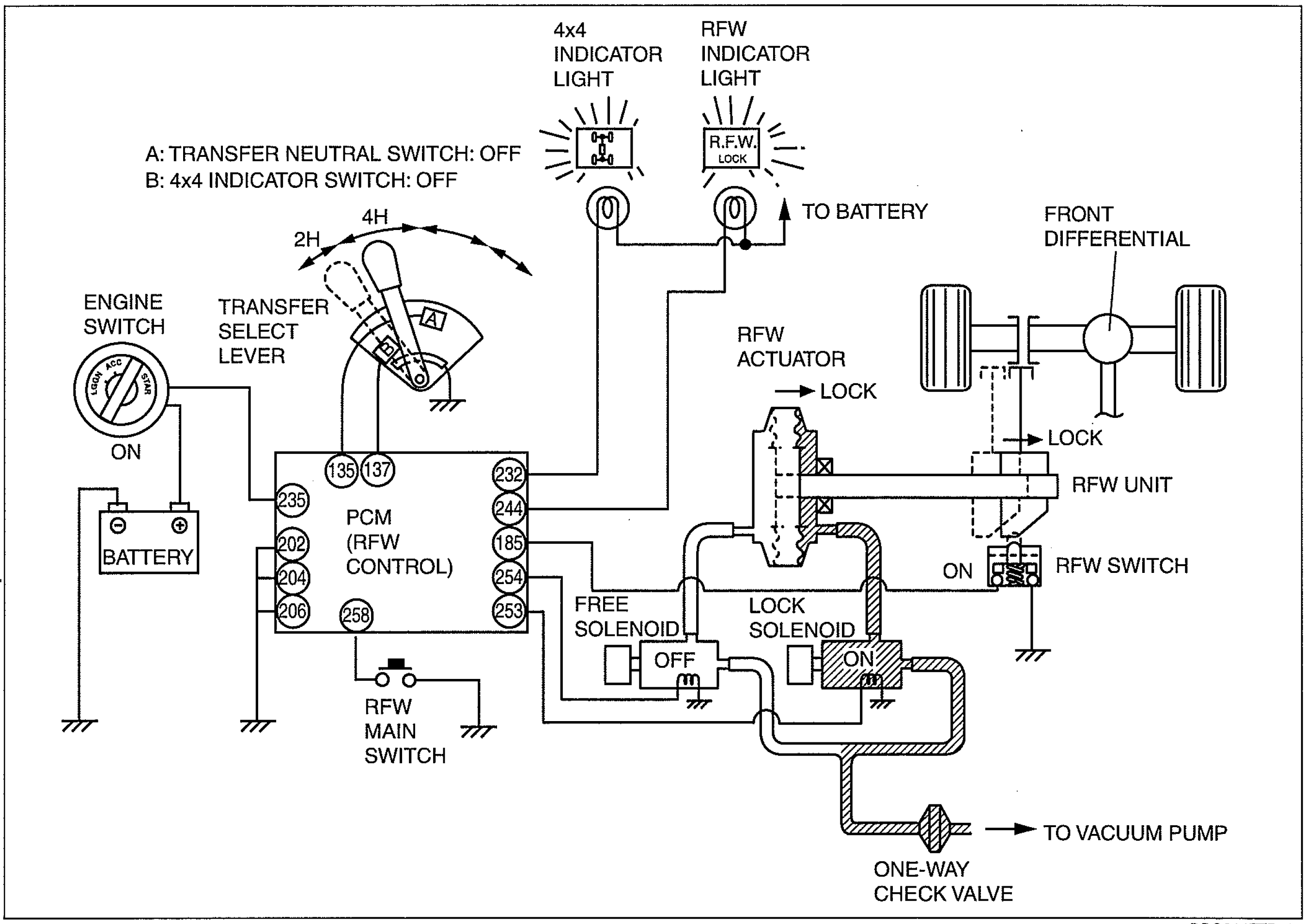
**Remote Freewheel Unit: LOCK**




DBG314ZTB006

- The RFW actuator pushes the sleeve over the output shaft, and the front differential output shaft and joint shaft are coupled together.
  If the sleeve and the output shaft are not aligned when actuated, the RFW actuator applies pressure to the sleeve unit it can slide into place.
  The rotation of the front propeller shaft is then transferred, via the front differential, to the front tires, and four-wheel-drive operation is possible.

## 2H (free) to 4H Selection

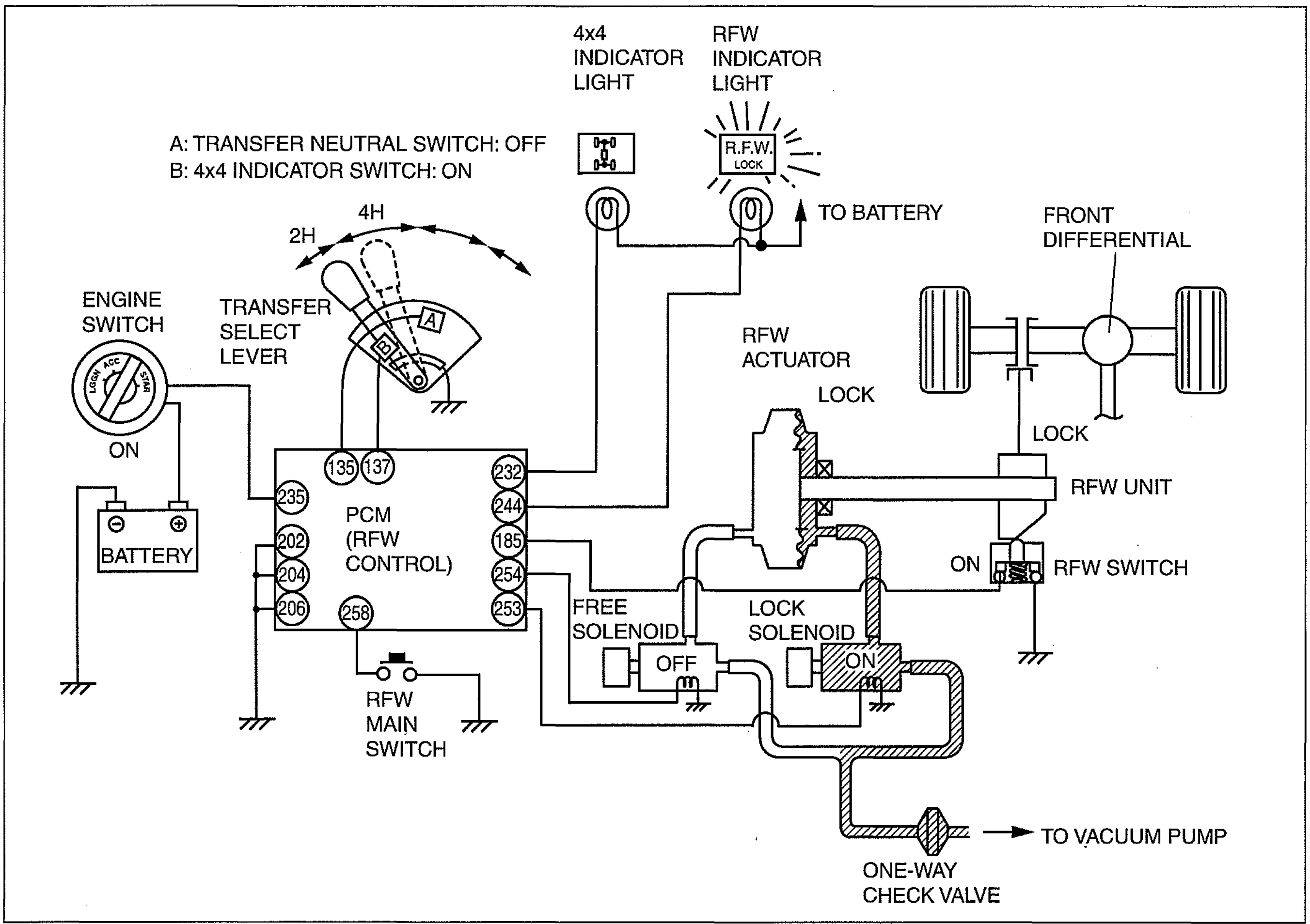



DBG314ZTB007

| Part name / Condition | Input: Transfer neutral switch | Input: 4x4 indicator switch | Input: RFW switch | Input: RFW main switch | Output: Lock solenoid | Output: Free solenoid | Output: 4x4 indicator light | Output: RFW indicator light | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|
| 2H-F | OFF | ON | OFF | - | OFF | ON | OFF | OFF | Free |
| 4H-$L_1$ | OFF | OFF | OFF | - | ON | OFF | ON | OFF | Lock |
| 4H-L | OFF | OFF | ON | - | ON | OFF | ON | ON | Lock |

- When the select lever is moved from 2H (2H-F in PCM) to 4H, the 4x4 indicator switch (in the transfer case switch) changes from ON to OFF. As a result, the PCM establishes a condition known as 4H-$L_1$, and the lock solenoid is switched ON, causing the actuator to move toward the lock side. At the same time, the 4x4 indicator light is illuminated.
  When the RFW unit is fully locked, the RFW switch is switched ON, and condition 4H-L is established in the PCM. The RFW indicator light illuminates to notify the driver that the RFW is locked.

### 4H to 2H (lock) Selection

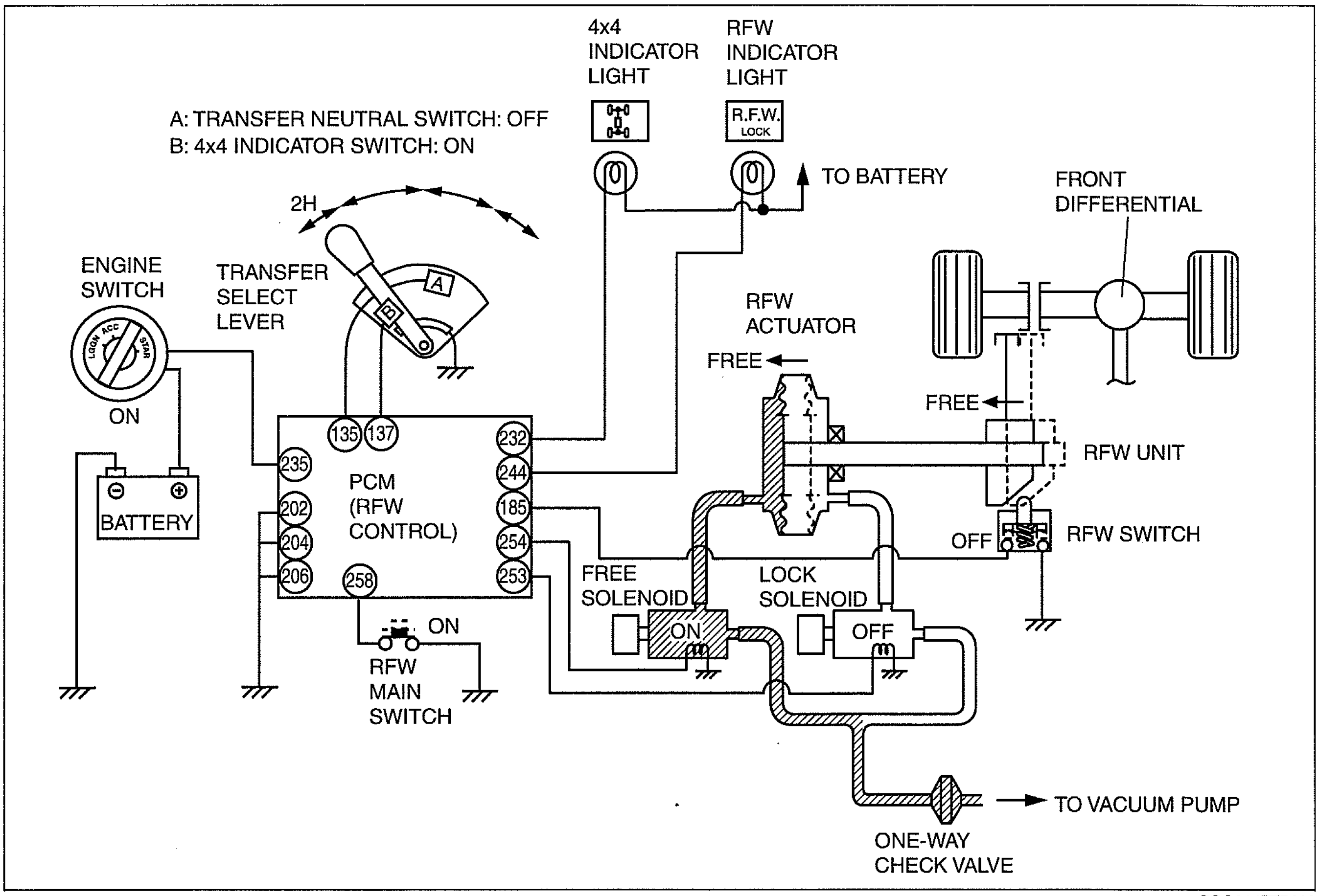



DBG314ZTB008

| Condition \ Part name | Input: Transfer neutral switch | Input: 4x4 indicator switch | Input: RFW switch | Input: RFW main switch | Output: Lock solenoid | Output: Free solenoid | Output: 4x4 indicator light | Output: RFW indicator light | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|
| 4H-F | OFF | OFF | ON | - | ON | OFF | ON | ON | Lock |
| 2H-L | OFF | ON | ON | OFF | ON | OFF | OFF | ON | Lock |

- When the select lever is moved from 4H (4H-L condition) to 2H, the 4x4 indicator switch (in the transfer case switch) changes from OFF to ON.
  If the RFW main switch is in the OFF position, condition 2H-L is established, the lock solenoid remains ON, and the RFW actuator is held at the lock position. At this time, the 4x4 indicator light is switched OFF.

## 2H (lock) to 2H (free) Selection

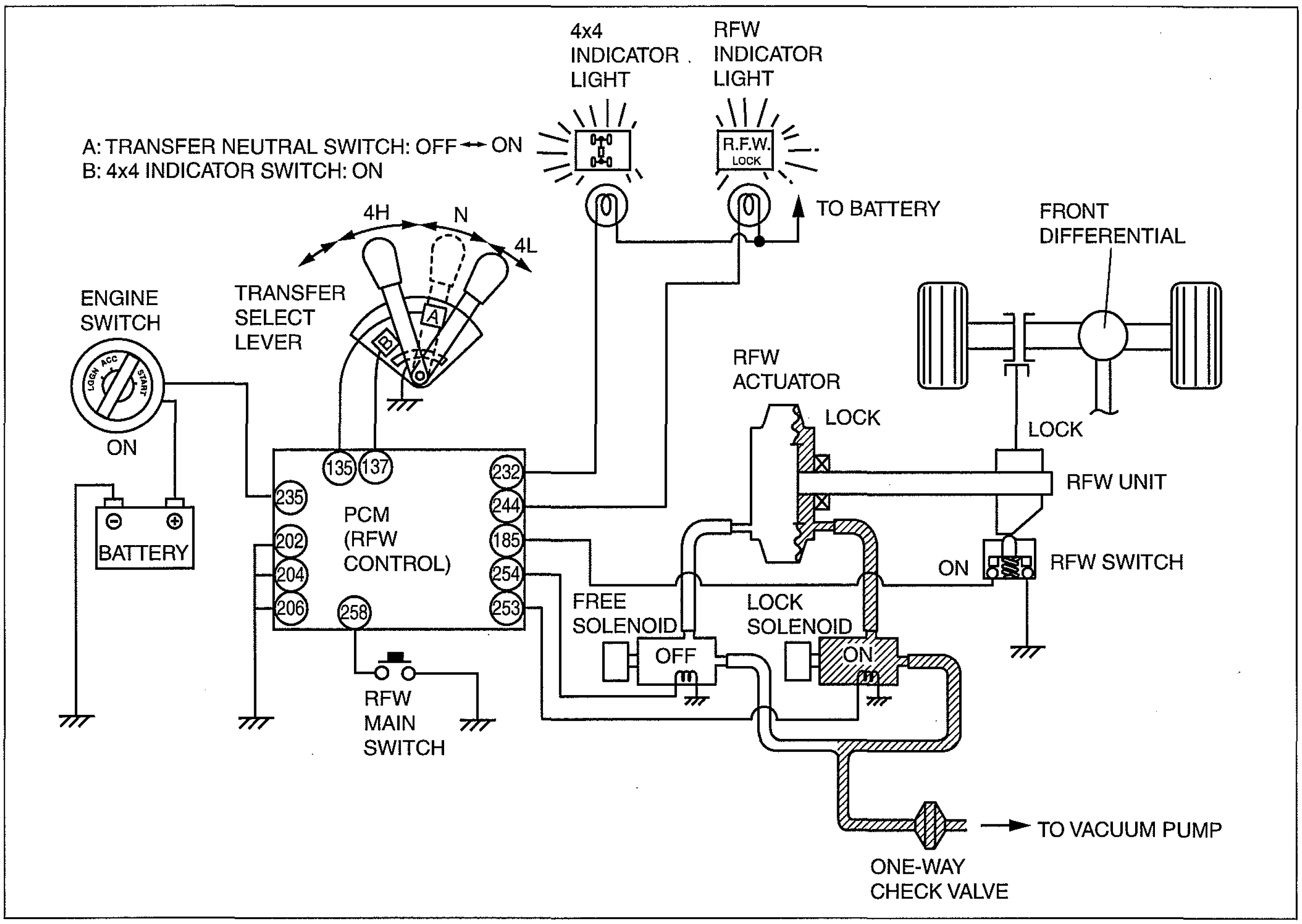



DBG314ZTB009

| Part name / Condition | Input | | | | Output | | | | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|
| | Transfer neutral switch | 4x4 indicator switch | RFW switch | RFW main switch | Lock solenoid | Free solenoid | 4x4 indicator light | RFW indicator light | |
| 2H-L | OFF | ON | ON | OFF | ON | OFF | OFF | ON | Lock |
| 2H-$F_1$ | OFF | ON | OFF | ON | OFF | ON | OFF | OFF | Free |
| 2H-F | OFF | ON | OFF | - | OFF | ON | OFF | OFF | Free |

- When the above condition (2H-L) the RFW main switch is pressed once (ON), condition 2H-$F_1$ is established, the RFW indicator light is switched OFF, and, same time, the free solenoid is switched ON, and the actuator is pulled to the free side. When the RFW unit becomes fully free, the RFW switch is turned OFF, and 2H-F condition is established.

### 4H to 4L or 4L to 4H Selection

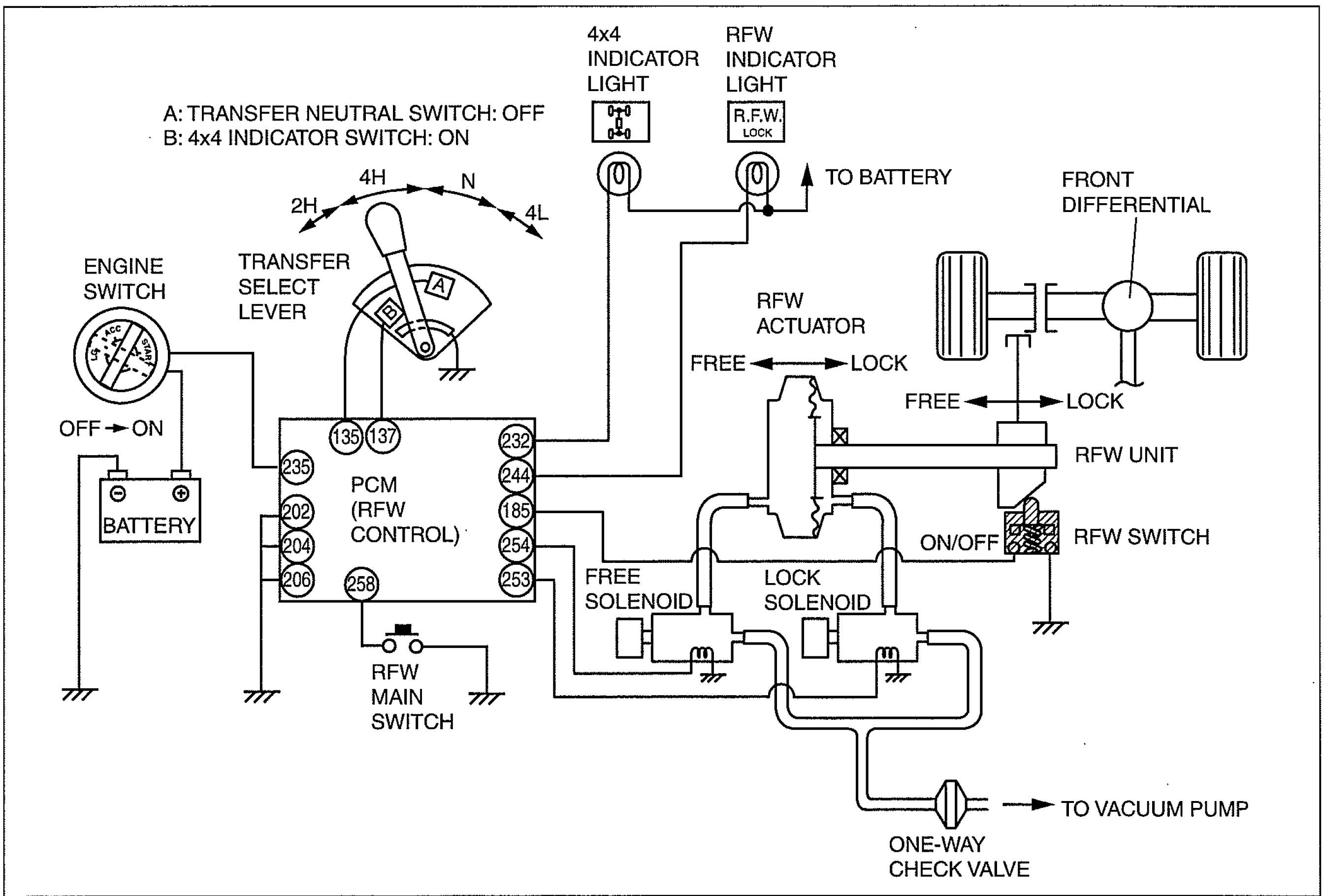



DBG314ZTB010

| Part name / Condition | Input | | | | Output | | | | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|
| | Transfer neutral switch | 4x4 indicator switch | RFW switch | RFW main switch | Lock solenoid | Free solenoid | 4x4 indicator light | RFW indicator light | |
| 4H-L | OFF | OFF | ON | - | ON | OFF | ON | ON | Lock |
| N | ON | OFF | ON | - | ON | OFF | OFF | ON | Lock |

- When the select lever is moved from 4H to 4L (both 4H-L in the PCM) or vice versa, there is a change to the neutral condition midway through the changeover thereby momentarily switching ON the transfer neutral switch (in the transfer case switch), and switching OFF the 4x4 indicator light.
  During 4H or 4L operation, the lock solenoid is ON (as a result of 4H-L condition), and the RFW actuator is held in the lock position. If the RFW main switch is pressed at this time, there is no effect upon the RFW actuator.

## Engine Switch Turned from OFF to ON




DBG314ZTB011

| Condition \ Part name | Input: Transfer neutral switch | Input: 4x4 indicator switch | Input: RFW switch | Input: RFW main switch | Output: Lock solenoid | Output: Free solenoid | Output: 4x4 indicator light | Output: RFW indicator light | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|
| 4H-L | OFF | OFF | ON | - | ON | OFF | ON | ON | Lock |
| 2H-L | OFF | ON | OFF | - | OFF | ON | OFF | OFF | Free |
| 2H-L | OFF | ON | OFF | OFF | OFF | ON | OFF | OFF | Free |

- When the ignition switch is turned from OFF to ON, the switches and solenoid are activated to provide the same RFW condition as when it was turned off.

# 03-18 4-WHEEL DRIVE [AT (5R55S)]



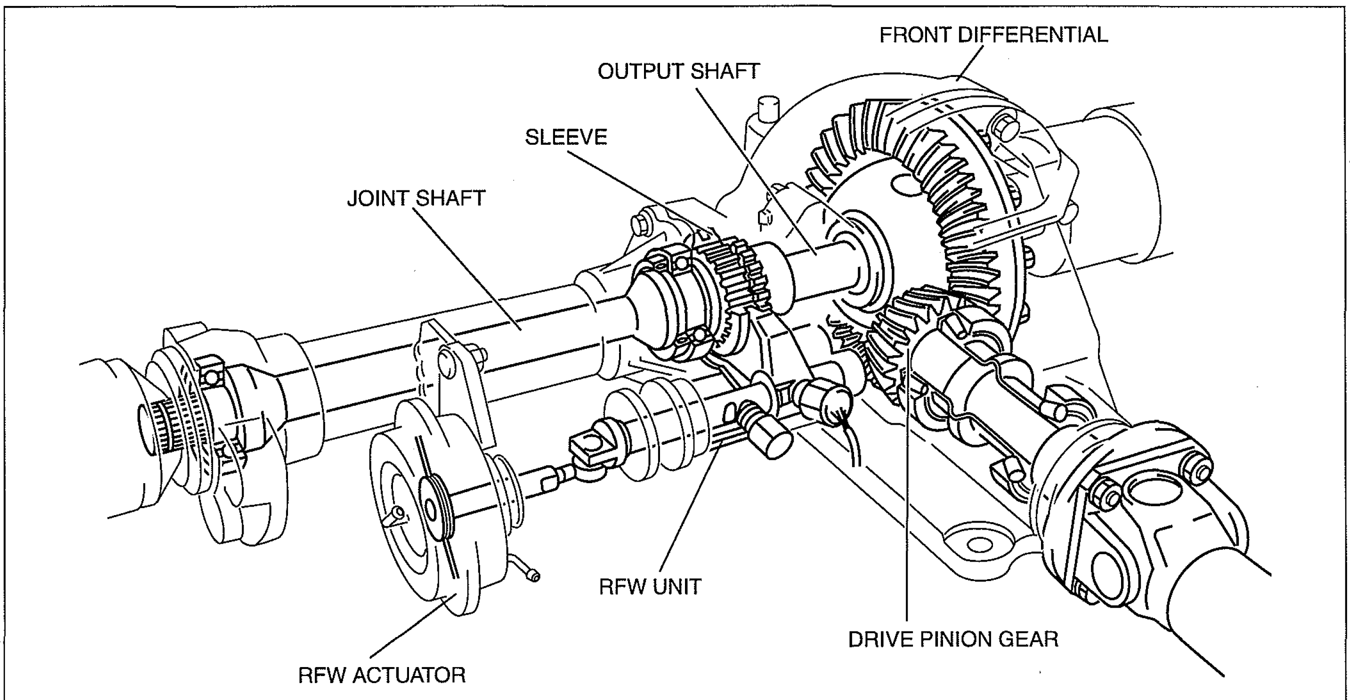
## 4x4 CONTROL SYSTEM OUTLINE [AT (5R55S)]

id0318d2164200

- A special 4x4 control module has been adopted for the 4x4 control system control, and it operates the transfer motor according to the signals from the speed sensor, 4x4 switch, and digital TR sensor to perform shifting between 2H, 4H and 4L.
- The 4x4 CM receives signals from the transfer case sensor and other switches. It then activates one of two solenoids, causing a vacuum from the vacuum pump to be applied to the RFW actuator, and switching the remote freewheel (RFW) unit to lock or free.
- To prevent reduced performance of the actuator as a result of a decrease in engine vacuum or vacuum pump, such as during acceleration or at high altitude operation, the actuation system includes a one-way check valve.
- The switching mechanism of the RFW unit is a mechanical dog clutch on the left side of the front differential.
- The RFW operation is controlled by 4x4 CM.

## 4x4 CONTROL SYSTEM CONSTRUCTION [AT (5R55S)]

id0318d2164100

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## 4x4 CONTROL SYSTEM COMPONENTS AND FUNCTIONS [AT (5R55S)]

id0318d2164000

### Electrical component

| Part name | | | Function |
|---|---|---|---|
| 4x4 CM | | | • Sends ON/OFF signals to lock and free solenoids and indicator lights according to signals from various switches |
| Input | Digital TR sensor | | • Detects selector lever N position |
| | 4x4 switch | | • Detects 4x4 switch 2H, 4H or 4L position |
| | Speed sensor | | • Detects output shaft speed |
| Output | Solenoid | | • Switched ON/OFF by electrical signals from 4x4 CM<br>• Regulates RFW unit "lock" or "free" |
| | | Lock | • Switched ON when "locked" |
| | | Free | • Switched ON when "free" |
| | Motor component (transfer case) | | • Sets transfer case operation mode (2H, 4H or 4L) |
| | Clutch coil (transfer case) | | • Operates when 4H or 4L selected |
| | 4x4 indicator light | | • Illuminates when 4H or 4L selected |
| | 4L indicator light | | • Illuminates when 4L selected |

### Mechanical component

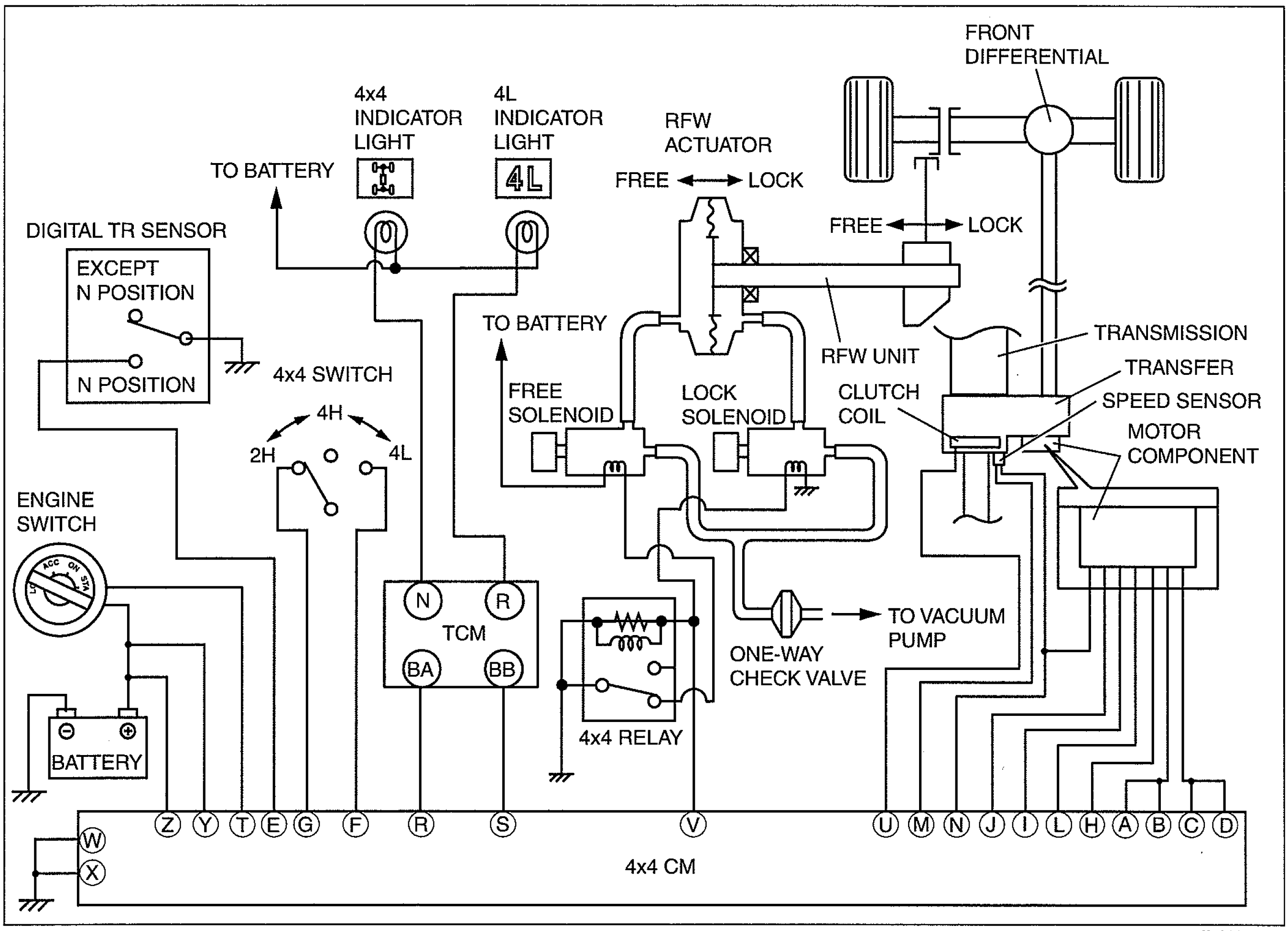
| Part name | Function |
|---|---|
| RFW unit | • Transmits front propeller shaft rotation to front tires |
| RFW actuator | • "Locks" or "free" RFW unit |
| One-way check valve | • Prevents leakage of vacuum |
| Motor component (transfer case) | • Motor operates to slide each shift fork |
| Clutch coil (transfer case) | • Magnetic force occur in clutch coil during operation to pull the lockup component |

### Relationship of Component and Function

| Part name / Function | Input | | | Output | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | **Digital TR sensor** | **4x4 switch** | **Speed sensor** | **Lock solenoid** | **Free solenoid** | **Motor component** | **Clutch coil** | **4x4 indicator light** | **4L indicator light** |
| Controlled by driver | x | x | | | | | | | |
| Controlled by 4x4 CM | | | x | x | x | x | x | x | x |

## 4x4 CONTROL SYSTEM SYSTEM DIAGRAM [AT (5R55S)]

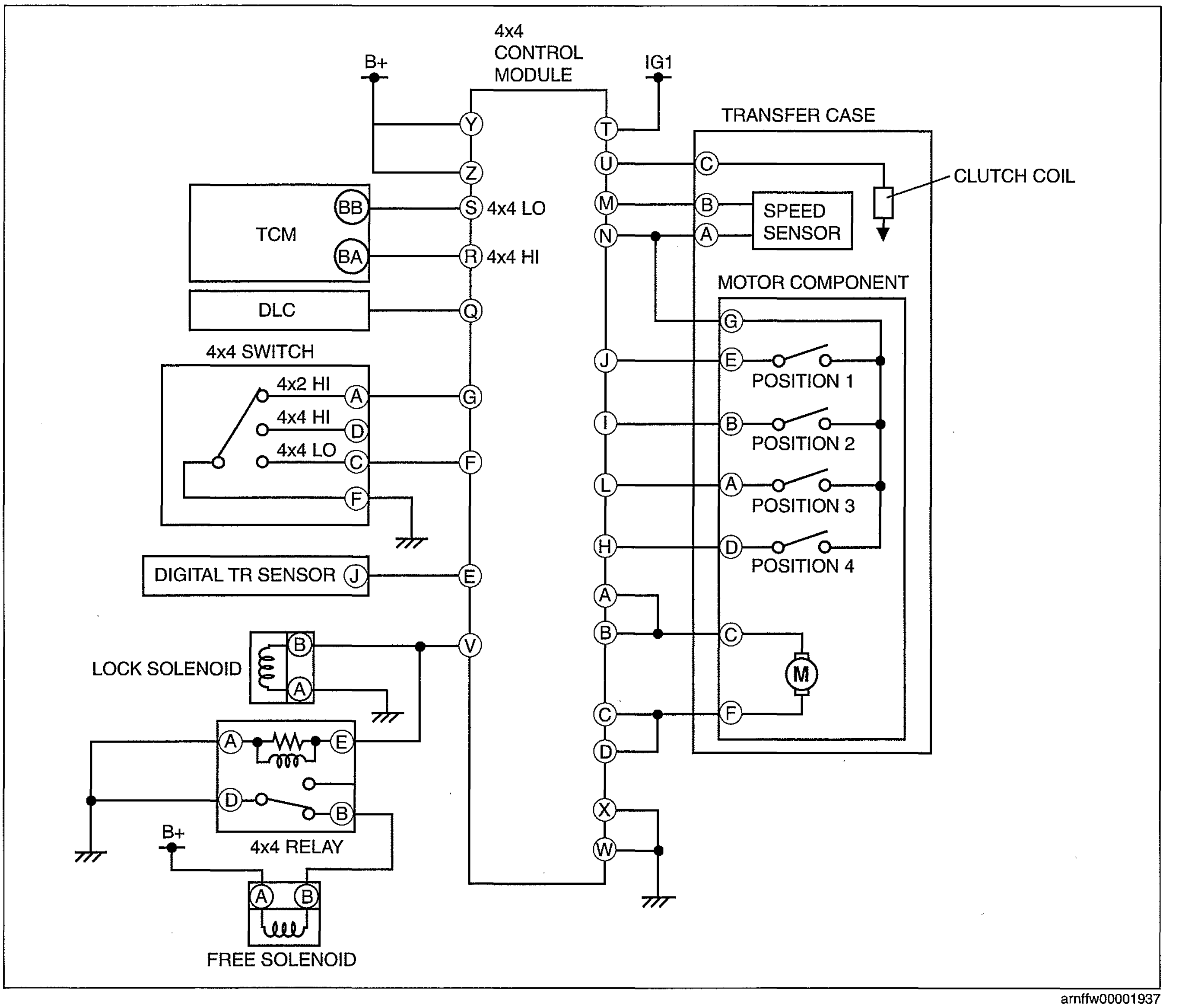
id0318d2163800




arnffn00000399

## 4x4 CONTROL SYSTEM SYSTEM WIRING DIAGRAM [AT (5R55S)]

id0318d2163900




arnffw00001937

## 4x4 CONTROL SYSTEM OPERATION [AT (5R55S)]

id0318d2165000

**Note**

- Shifting from 2H to 4H or 4H to 2H can be performed whether the vehicle is stopped or driving.

### 2H to 4H Selection

- When the 4x4 switch is switched from 2H to 4H, the clutch coil and motor in the transfer operate, the lockup shift fork slides to establish the 4H condition and the RFW unit locks automatically. At this time, the 4x4 indicator light illuminates to inform the driver that the vehicle is in four-wheel-drive mode.

### 4H to 2H Selection

- When the 4x4 switch is switched from 4H to 2H, the clutch coil in the transfer stops operation but the motor operates to slide the lockup shift fork and establish the 2H condition. Also, the RFW unit unlocks automatically. At this time, the 4x4 indicator light turns off to inform the driver that the vehicle is in 4x2 mode.

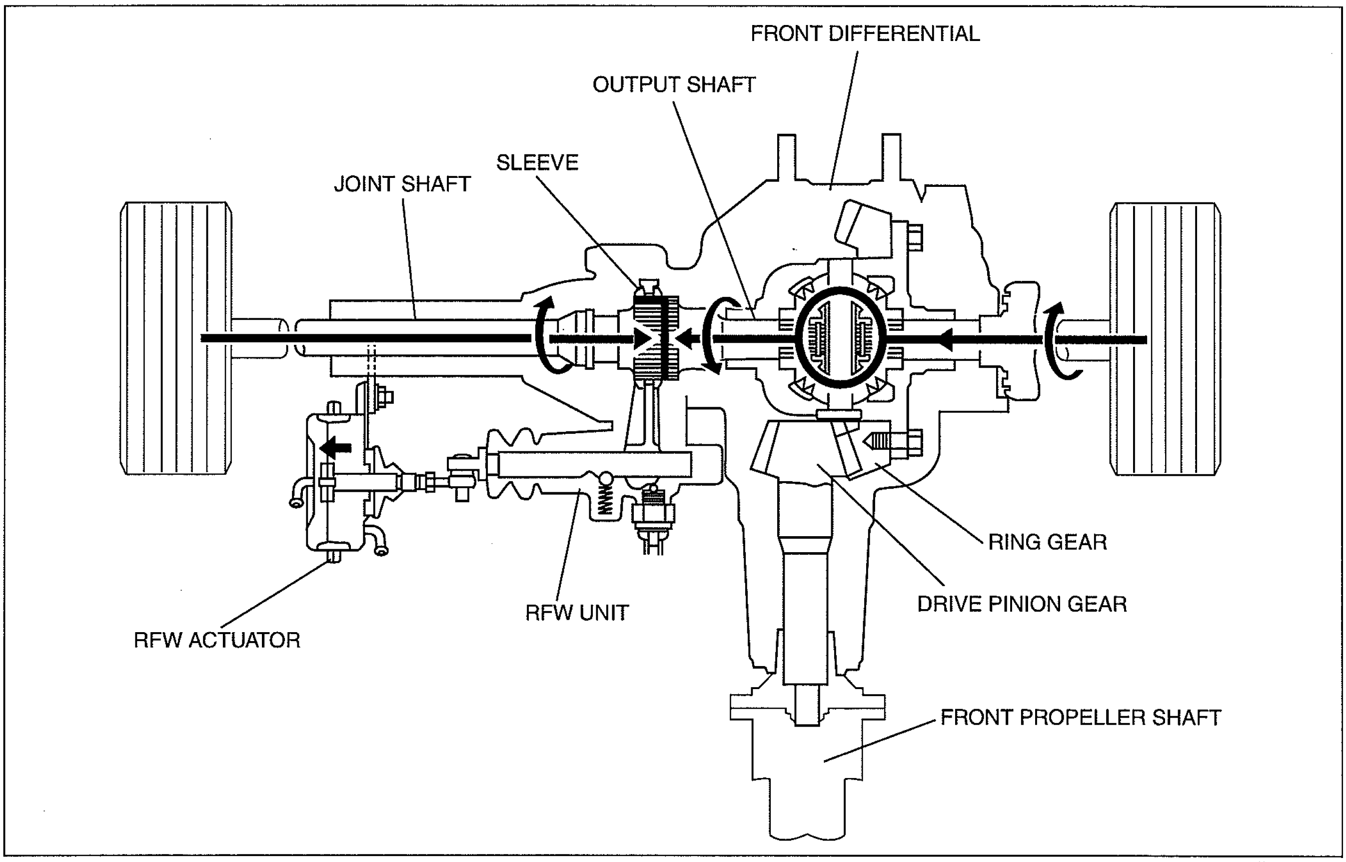
### 4H to 4L Selection

- Shifting from 4H to 4L can be performed only when the vehicle is stopped with the selector lever in the N position. When the 4x4 switch is switched from 4H to 4L, the motor in the transfer operates and the reduction shift fork slides to establish the 4L condition. At this time, the RFW unit remains locked and the 4x4 indicator light remains illuminated and, in addition, the 4L indicator light illuminates.

**4L to 4H Selection**

- Shifting from 4L to 4H can be performed only when the vehicle is stopped with the selector lever in the N position. When the 4x4 switch is switched from 4L to 4H, the motor in the transfer operates and the reduction shift fork slides to establish the 4H condition. At this time, the RFW unit remains locked and the 4x4 indicator light remains illuminated, but the 4L indicator light turns off.

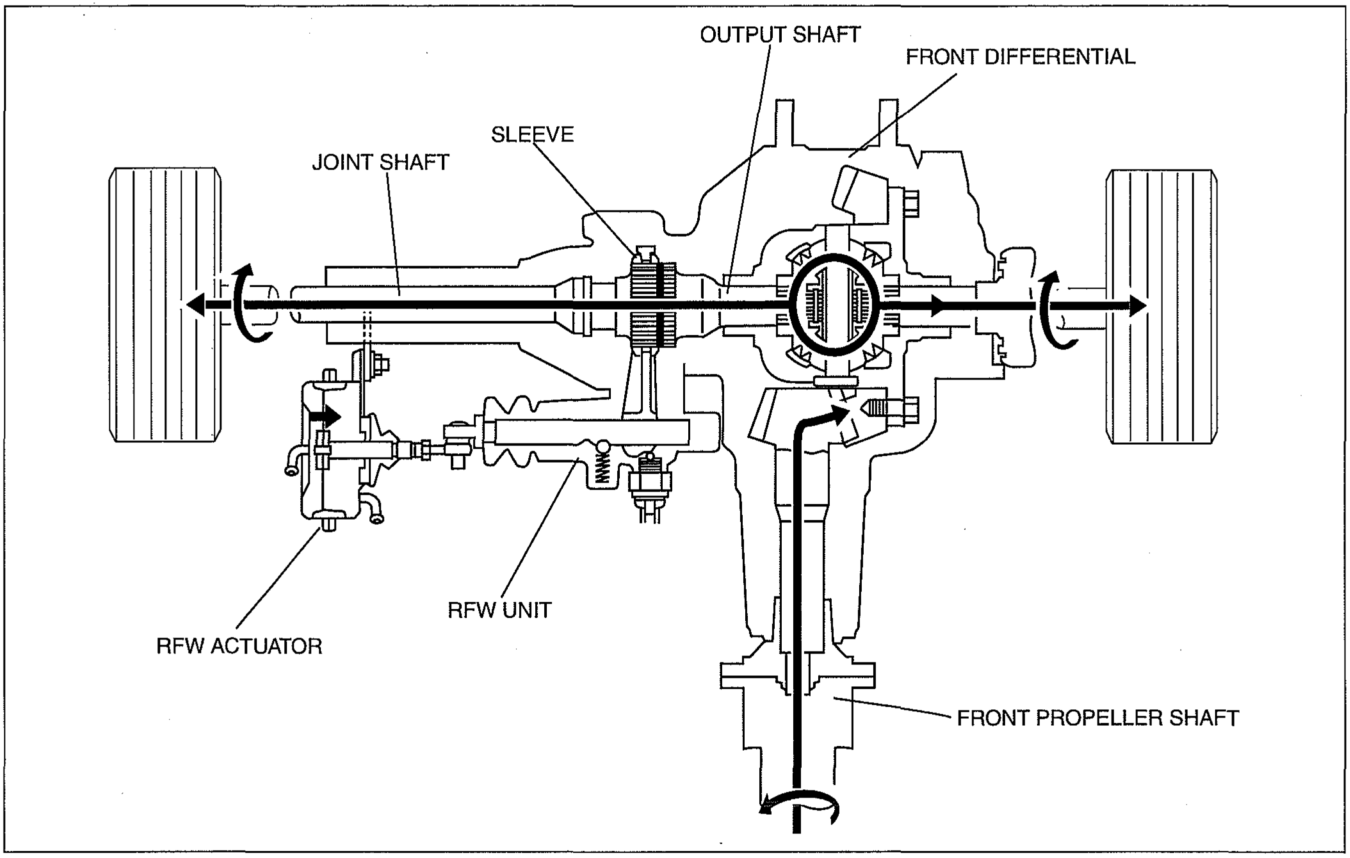
**Remote Freewheel Unit: FREE**




arnffn00000029

- The RFW actuator pulls the sleeve away from the front differential output shaft, allowing the front differential to rotate freely.
  The rotation of the right front tire is absorbed by the front differential, with the result that the ring gear, drive pinion gear, and front propeller shaft do not turn.

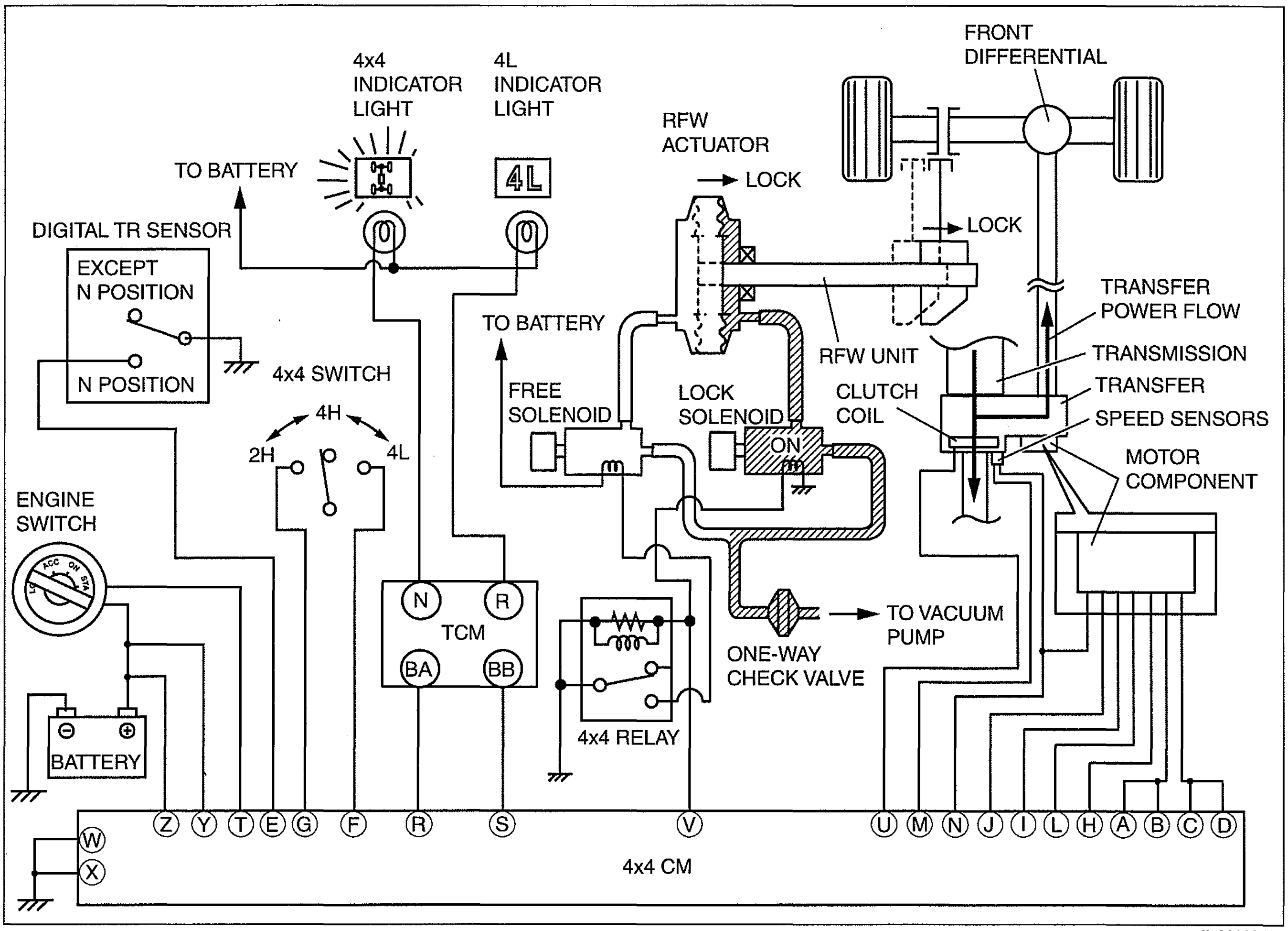
**Remote Freewheel Unit: LOCK**




arnffn00000030

- The RFW actuator pushes the sleeve over the output shaft, and the front differential output shaft and joint shaft are coupled together.
  If the sleeve and the output shaft are not aligned when actuated, the RFW actuator applies pressure to the sleeve unit it can slide into place.
  The rotation of the front propeller shaft is then transferred, via the front differential, to the front tires, and four-wheel-drive operation is possible.

## 2H to 4H Selection

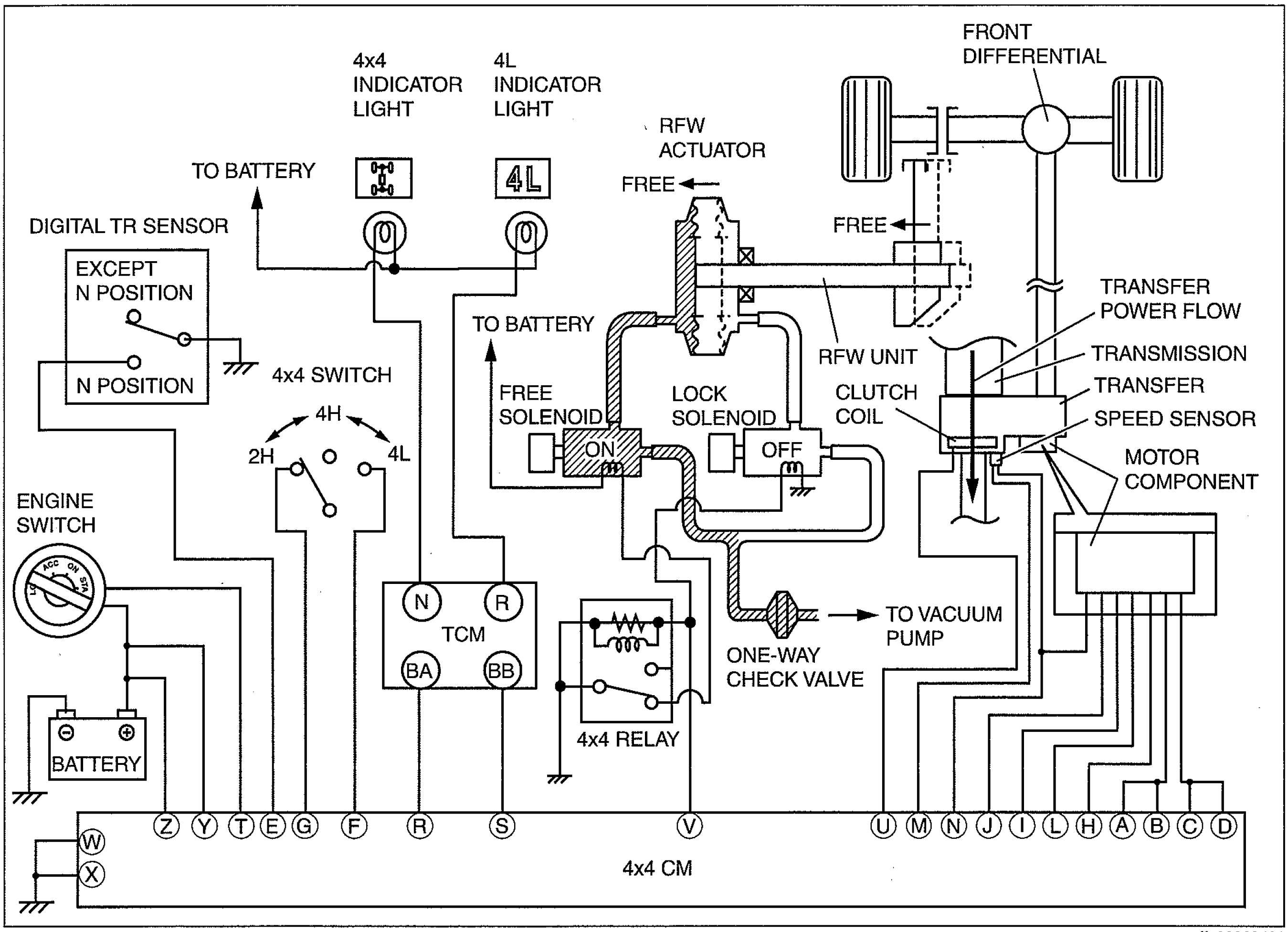



arnffn00000400

| Part name / Condition | Input | | | Output | | | | | | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|---|
| | Digital TR sensor | 4x4 switch | Speed sensor | Lock solenoid | Free solenoid | Motor component | Clutch coil | 4x4 indicator light | 4L indicator light | |
| 2H | - | 2H | - | OFF | ON | 2H | OFF | OFF | OFF | Free |
| 4H | - | 4H | - | ON | OFF | 4H | ON | ON | OFF | Lock |

- When the 4x4 switch is switched from 2H to 4H, the clutch coil and motor in the transfer operate, the lockup shift fork slides to establish the 4H condition and the RFW unit locks automatically. At this time, the 4x4 indicator light illuminates to inform the driver that the vehicle is in four-wheel-drive mode.

## 4H to 2H Selection

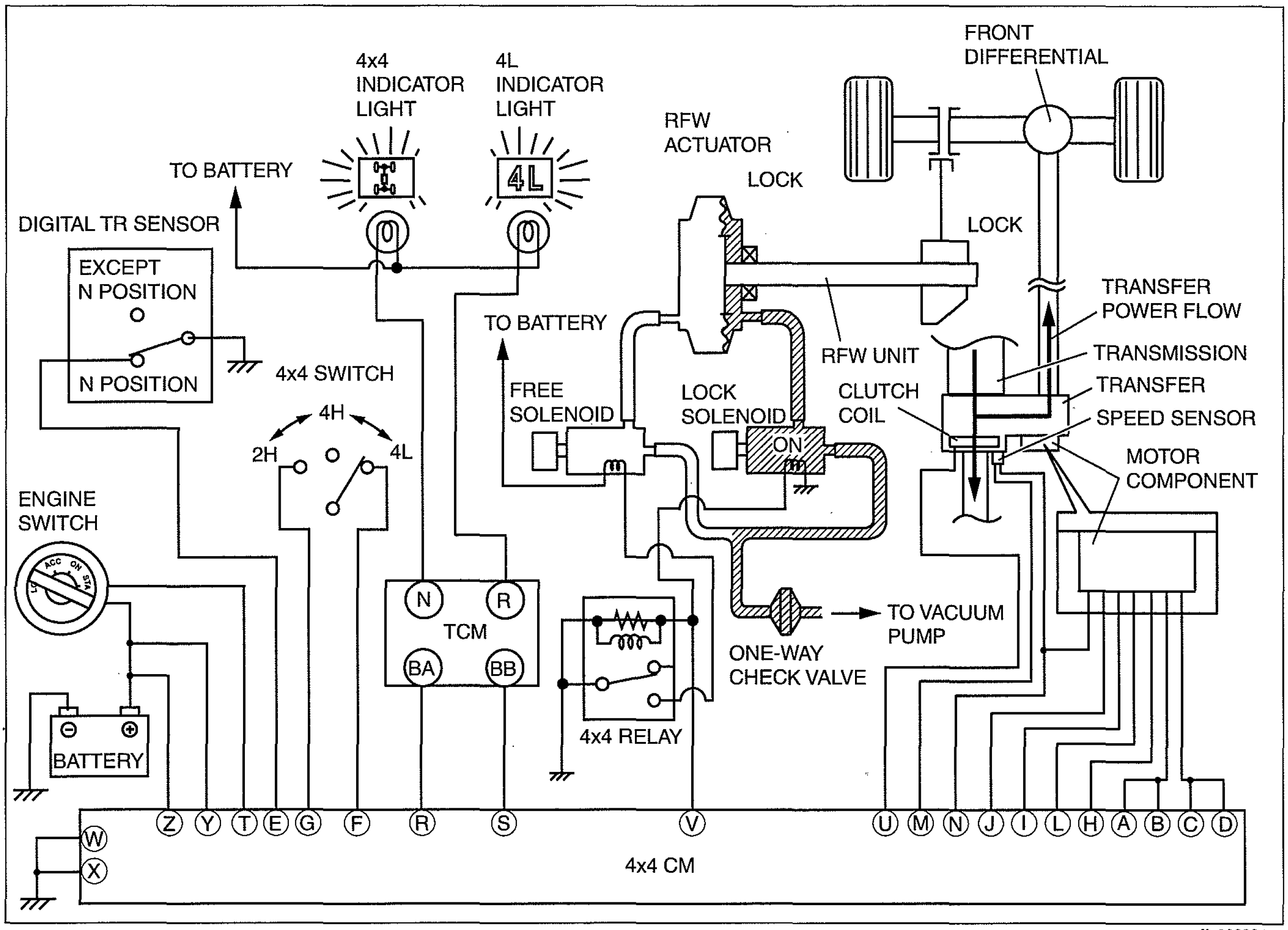



arnffn00000401

| Condition \ Part name | Input | | | Output | | | | | | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|---|
| | Digital TR sensor | 4x4 switch | Speed sensor | Lock solenoid | Free solenoid | Motor component | Clutch coil | 4x4 indicator light | 4L indicator light | |
| 4H | - | 4H | - | ON | OFF | 4H | ON | ON | OFF | Lock |
| 2H | - | 2H | - | OFF | ON | 2H | OFF | OFF | OFF | Free |

- When the 4x4 switch is switched from 4H to 2H, the clutch coil in the transfer stops operation but the motor operates to slide the lockup shift fork and establish the 2H condition. Also, the RFW unit unlocks automatically. At this time, the 4x4 indicator light turns off to inform the driver that the vehicle is in 4x2 mode.

## 4H to 4L Selection

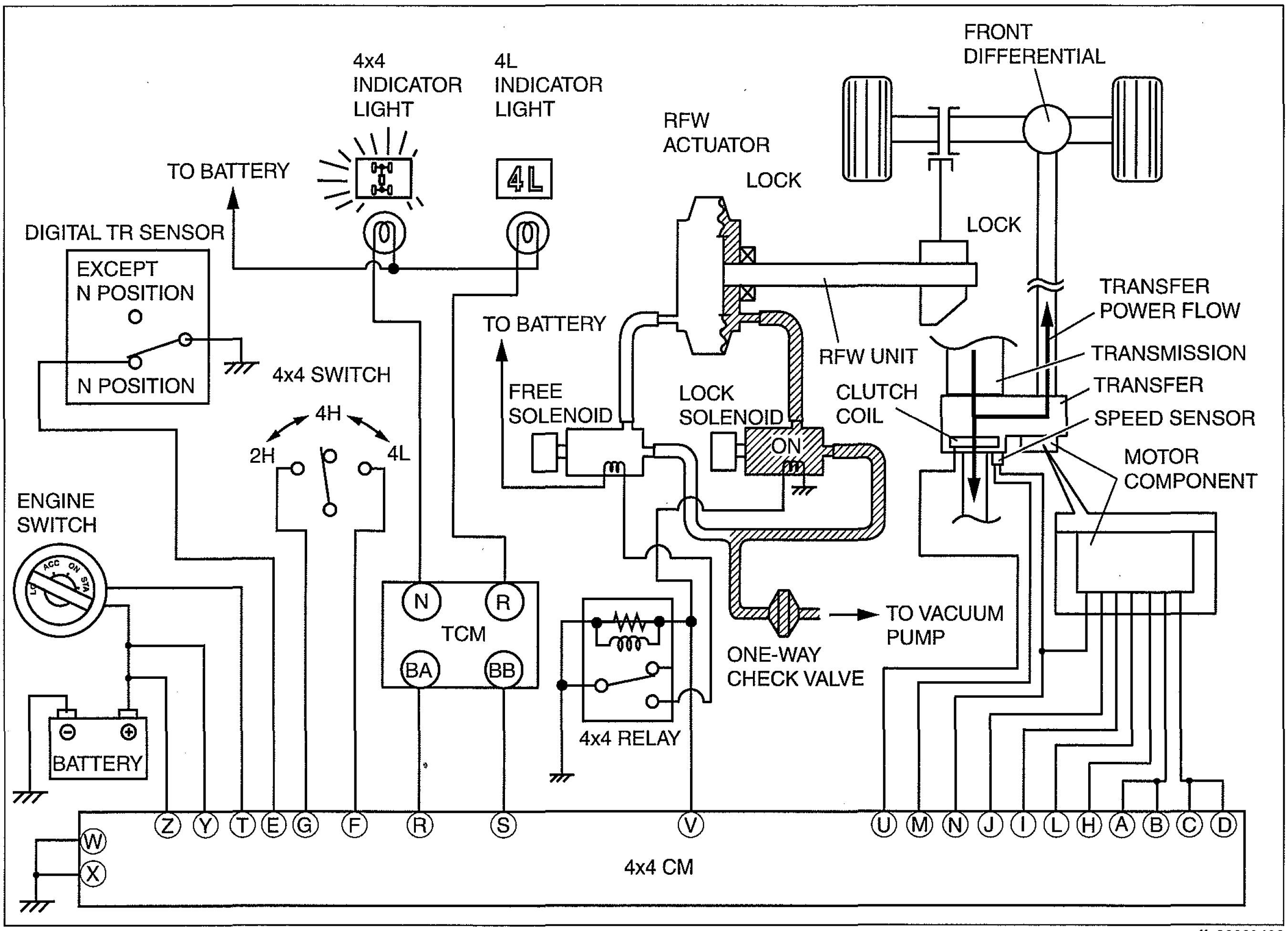


arnffn00000402

| Part name / Condition | Input | | | Output | | | | | | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|---|
| | Digital TR sensor | 4x4 switch | Speed sensor | Lock solenoid | Free solenoid | Motor component | Clutch coil | 4x4 indicator light | 4L indicator light | |
| 4H | N | 4H | vehicle stop | ON | OFF | 4H | ON | ON | OFF | Lock |
| 4L | N | 4L | vehicle stop | ON | OFF | 4L | ON | ON | ON | Lock |

- Shifting from 4H to 4L can be performed only when the vehicle is stopped with the selector lever in the N position. When the 4x4 switch is switched from 4H to 4L, the motor in the transfer operates and the reduction shift fork slides to establish the 4L condition. At this time, the RFW unit remains locked and the 4x4 indicator light remains illuminated and, in addition, the 4L indicator light illuminates.

### 4L to 4H Selection



amffn00000403

| Part name / Condition | Input | | | Output | | | | | | RFW actuator Position |
|---|---|---|---|---|---|---|---|---|---|---|
| | Digital TR sensor | 4x4 switch | Speed sensor | Lock solenoid | Free solenoid | Motor compon ent | Clutch coil | 4x4 indicato r light | 4L indicato r light | |
| 4L | N | 4L | vehicle stop | ON | OFF | 4L | ON | ON | ON | Lock |
| 4H | N | 4H | vehicle stop | ON | OFF | 4H | ON | ON | OFF | Lock |

- Shifting from 4L to 4H can be performed only when the vehicle is stopped with the selector lever in the N position. When the 4x4 switch is switched from 4L to 4H, the motor in the transfer operates and the reduction shift fork slides to establish the 4H condition. At this time, the RFW unit remains locked and the 4x4 indicator light remains illuminated, but the 4L indicator light turns off.